Reconstructing founder haplotypes, pedigrees and recombination from low-coverage sequencing of experimental crosses

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Abstract

Low-coverage whole-genome sequencing makes it affordable to sequence every member of an experimental cross, but leaves each individual's genotypes uncertain, and in many crosses the founder haplotypes, the number of founders and the pedigree are themselves unknown. We describe a method that starts from nothing more than per-sample allele depths and a genetic map. It reconstructs the founder haplotypes of the cross locally in marker blocks, without imputing missing data, and assembles them along each chromosome with an explicit rule for the number of founders. It then paints every sample as a diploid mosaic of those founders, infers the pedigree with every sequenced sample a candidate parent of every other, refines phase conditional on the inferred pedigree and estimates recombination maps from the resulting meiosis chains. No sample metadata, sex, cohort or known parent enters inference; parents that were not sequenced and relationships that the data cannot resolve are reported as such rather than forced. Every stage is designed to stay polynomial in the number of founder haplotypes up to K = 30, and the implementation gives identical results whatever the number of processor cores. TO FILL: one or two sentences of headline results from the simulation matrix and the TroMau cross, once the 4.0 validation matrix is scored. The software is open source (MIT licence).

1 Introduction

Experimental crosses remain one of the most direct ways to connect genetic variation with phenotype, to measure recombination and to study how divergent genomes interact in hybrids. Their power rests on a simple structure: every chromosome of a descendant is a mosaic of a limited number of founder haplotypes, joined at crossovers that occurred in the meioses of the pedigree. Once the founder haplotypes and the mosaic of each individual are known, genotypes can be completed from very sparse data, crossovers can be counted, and the pedigree itself can be read off the shared segments.

Methods that exploit this structure have a long history. Hidden Markov models of inheritance along a chromosome in a known pedigree (Lander and Green 1987), and of founder ancestry in multiparent populations (Mott et al. 2000; Broman et al. 2019), reconstruct each individual's mosaic when the founder genotypes and the design of the cross are known. Haplotype-copying models (Li and Stephens 2003) describe a sample as a mosaic of other haplotypes and underlie most modern phasing and imputation methods, including imputation from low-coverage sequence with large reference panels (Rubinacci et al. 2021) and, without any reference panel, imputation from a small number of inferred ancestral haplotypes (Davies et al. 2016). Pedigree-aware phasing (O'Connell et al. 2014) and linkage mapping from low-coverage data (Rastas 2017) exploit known relationships. Pairwise relatedness can be estimated directly from genotype likelihoods (Thompson 1975; Korneliussen and Moltke 2015), and parentage and sibship can be reconstructed from called SNP genotypes (Jones and Wang 2010; Huisman 2017).

Crosses in non-model organisms often fall between these assumptions. In the cichlid crosses that motivated this work, drawn from the adaptive radiation of Lake Malawi (Malinsky et al. 2018), individuals are sequenced at a few-fold coverage, so that genotype calls at single sites are unreliable. Not every founder individual is sequenced, and those that are sequenced need not be parents of the others. The number of distinct founder haplotypes is not known in advance and varies along the genome, because founders may share haplotypes locally. Breeding records may be incomplete, and cohort or generation

labels describe when an animal was sampled rather than who its parents were. A method that conditions on such labels, or on a declared design, risks reproducing errors in the records as apparent biological signal.

Here we describe a method, released as version 4.0 of the software package `haplotype_reconstruction`, that addresses this setting jointly. It reads only a multi-sample VCF or BCF file with per-sample allele depths (Danecek et al. 2011, 2021) and a genetic map; sample names are treated as identifiers and nothing else about a sample enters inference. From these inputs it

1. models read evidence and genotype likelihoods, with data-driven site and sample quality control;
2. estimates pairwise identity by descent and full-sib families from the genotype likelihoods of the whole genome;
3. discovers founder haplotypes in short marker blocks with a missing-aware model, and selects local founder panels using feedback from the chromosome-scale assembly;
4. assembles local panels into chromosome-scale founder haplotypes, deciding the number of founders with an explicit count rule and completing missing founder alleles jointly;
5. paints every sample as a diploid mosaic of the founders, and refits individual founder alleles with those paintings held fixed;
6. infers the pedigree, with every sequenced sample a candidate parent of every other, and releases parent calls in evidence tiers that keep missing, ambiguous and unresolved parents explicit;
7. refines sample phase conditional on the inferred pedigree; and
8. estimates recombination maps from the meiosis chains of the final phase.

Three principles run through the design. First, the inputs are the data alone: the direction of a parent–offspring relationship comes from the genetic evidence, never from a cohort label or a sex annotation, and the release rules prefer an explicit unresolved outcome to a forced call. Second, the method is meant to remain usable as crosses grow: the documented scale envelope targets up to N = 2,500 samples on bounded genome crops and K = 30 founder haplotypes, nothing is exponential in K, and every stage is held to at most cubic work in K apart from a small number of documented exemptions and exceptions. Third, results are reproducible: all ties are resolved by declared rules, random draws come from named streams, and the released outputs do not depend on the number of processor cores used.

We describe the models and decision rules of each stage, the simulation framework used to measure accuracy against recorded truth, and the computational design, and we report the measured scaling of every stage. TO FILL: summary of the accuracy results on simulated crosses and of the analysis of the TroMau cross.

2 Methods

2.1 Overview and notation

The method is a single pipeline of stages connected by a fixed dependency graph (Figure 1). Stages that act within one chromosome (contig) run on each contig independently and concurrently; two stages, relatedness and pedigree, pool information across the genome. Relatedness consumes the evidence of every contig and nothing else, so that families are estimated from genotype likelihoods alone, without founders, painting, pedigree, phase or labels; it runs alongside the per-contig reconstruction, which does not depend on it. The pedigree consumes the relatedness families together with the paintings and genotype likelihoods of all contigs. Family phase and recombination act per contig after the pedigree is fixed, and never feed back into founder or pedigree inference. A genetic map,

in centimorgans, gives the genetic distances used by relatedness and by every stage from local selection onwards; where no map is given, or beyond its last position, a uniform fallback rate is used. Founder discovery works within short blocks and uses no map.

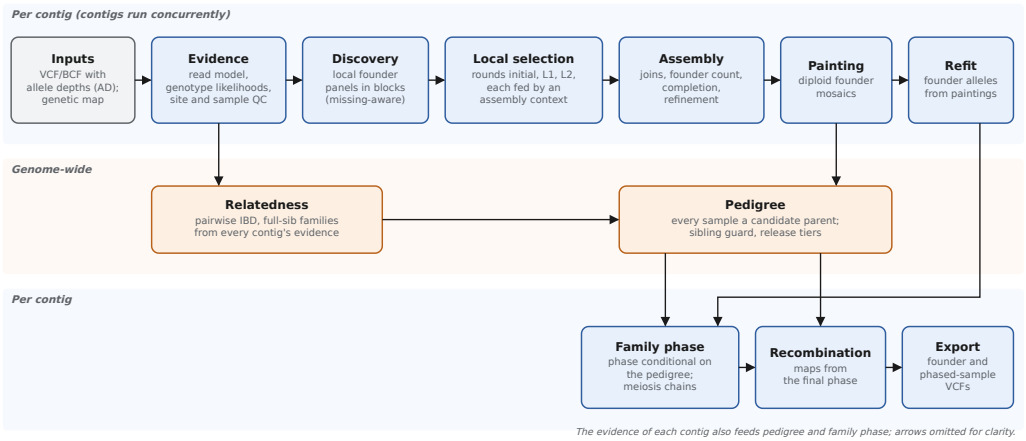


Figure 1. The stage graph of release 4.0. Each per-contig chain (blue) runs independently for every contig. Relatedness consumes only the evidence of all contigs and runs alongside the per-contig reconstruction; pedigree inference pools the relatedness families, paintings and genotype likelihoods of all contigs (orange). Local selection proceeds in three rounds (initial, L1 and L2); the second and third use an assembly context built from the preceding round. Family phase and recombination are conditional on the inferred pedigree and do not feed back upstream. The genetic map enters relatedness and every stage from local selection onwards.

Throughout, N is the number of sequenced samples, indexed by i; K the number of founder haplotypes, indexed by k; and L the number of biallelic sites on a contig, indexed by j, of which M may be selected for a given computation. B denotes the number of bins or blocks into which a contig is divided. Painting, pedigree and phase models use $S \le K+1$ states per haplotype: one per class of founder rows that are identical over the painted sites, and one background state. For sample i at site j, r_{ij} and a_{ij} are the reference and alternate read counts and $d_{ij} = r_{ij} + a_{ij}$ the depth; $g \in \{0,1,2\}$ is the alternate-allele dosage of a diploid genotype; and $h_{kj} \in \{0,1\}$ is the allele carried by founder k. Where a stage needs the probability of recombination in a meiosis over a genetic distance of x morgans, it uses the Haldane recombination fraction

$$r_H(x) = \tfrac{1}{2}(1 - e^{-2x}) = -\tfrac{1}{2}\operatorname{expm1}(-2x),$$

evaluated in this form for numerical accuracy at short distances (Haldane 1919). The copying models of local selection, assembly and refinement instead convert map distance into switch probabilities of their own, described with each stage. Further symbols are defined where they are used.

No sample metadata. Sample names identify rows of the input and carry no other meaning. No generation, cohort, sex, breeding record or known-parent label enters any stage. Every sequenced sample is a candidate parent of every other, parent slots are unordered and carry no sex, and the direction of a parent–offspring relationship is inferred from the data. Individuals that are not part of the cross must be removed from the input, because the method cannot exclude them by label.

Uncertainty is released, not hidden. Founder alleles that the data do not determine remain missing rather than imputed; painting, pedigree and phase calls carry their support; and parents that were not sequenced, are ambiguous between candidates, or cannot be resolved are reported as such. Every search in the pipeline has a declared order and budget, and every tie between numerically indistinguishable alternatives is resolved by a declared rule (Implementation, below).

The sections below give the model and decision rules of each stage. The Supplementary Information (sections S1–S6) gives their finer rules with the default value of every named parameter, and codes there such as EC05 or AJ17 point to the corresponding unit of the software's methods documentation.

2.2 Read evidence, quality control and relatedness

2.2.1 Read evidence and genotype likelihoods

Observations. The input is an indexed multi-sample VCF or BCF file (Danecek et al. 2011, 2021) with per-sample allele depths (FORMAT/AD). A record is admitted only if it is a biallelic SNP with single-nucleotide REF and ALT alleles. When two or more admitted records share a position, all of them are removed. Missing AD values count as zero reads; input FILTER flags and called genotypes do not enter as evidence. The observation for sample i at site j is the read pair (r_{ij}, a_{ij}) with depth $d_{ij} = r_{ij} + a_{ij}$, and a cell with $d_{ij} = 0$ is unobserved. Samples are ordered canonically by name, and names are identifiers only: no generation, cohort, sex or parentage information enters any stage.

Read model. Given the genotype g, the ALT count of a cell follows

$$\Pr(a_{ij} \mid d_{ij}, g) \propto \begin{cases} e_i^{a_{ij}} (1-e_i)^{r_{ij}}, & g = 0, \\ \dfrac{\mathrm{B}(a_{ij} + \kappa q_i,\ r_{ij} + \kappa(1-q_i))}{\mathrm{B}(\kappa q_i,\ \kappa(1-q_i))}, & g = 1, \\ (1-e_i)^{a_{ij}} e_i^{r_{ij}}, & g = 2, \end{cases} \qquad q_i = e_i + (1-2e_i) b_i,$$

where the binomial coefficient common to the three genotypes is omitted. Here e_i is a homozygote's wrong-allele read probability, and the heterozygote ALT probability q_i combines it with an allelic balance b_i. Heterozygote reads are overdispersed through a beta-binomial law with concentration κ, and $\kappa \to \infty$ recovers the binomial heterozygote. Homozygotes are binomial in every model.

Calibration and model choice. The read model is fitted per contig, from its allele depths alone, on the sites kept by the quality filter below. Three nested candidates are fitted:

1. a shared binomial model, with one e and one q for all samples and $\kappa = \infty$, fitted by expectation–maximisation (Dempster, Laird, and Rubin 1977) from several starts;
2. a sample binomial model, with per-sample effects $\operatorname{logit} e_i$ and $\operatorname{logit} b_i$ under normal priors whose mean and between-sample variance are estimated by empirical Bayes, each sample's sampling variance coming from a diagonal Laplace approximation;
3. a sample beta-binomial model, which refits these effects jointly with one shared κ under the same priors.

Each sample's genotype proportions are fitted as nuisance parameters. They are never returned as genotype priors, so the genotype likelihoods carry read evidence only.

Fitting and model choice use different physical folds. A site at position x_j (bp) falls in fold $\lfloor x_j / 10^6 \rfloor \bmod 4$ (1-Mb windows). Folds 0 and 1 train, folds 2 and 3 are held out, and only cells with $1 \le d_{ij} \le 48$ enter fitting and scoring. Held-out log predictive likelihoods are summed over each held-out stretch, which is the held-out half of one four-window cycle. Candidates are compared in order of complexity. Each one replaces the currently accepted model when its total held-out gain Δ is positive, at least four stretches contribute, and

$$z = \frac{\Delta}{\sqrt{n_h\, s_\Delta^2}} \ge 2,$$

where s_Δ^2 is the sample variance of the n_h stretch gains. The standard error comes from whole stretches, not single sites, because sites linked through ancestry or sharing reads are not independent observations. A fixed binomial model ($e = 0.02$, $q = 1/2$) is used instead

when calibration is disabled, a fold has no admitted cells, the shared model cannot be fitted or resolved to one optimum, or its held-out score is not finite.

Genotype likelihoods. The selected model is applied to every cell of the contig, at any depth and without a genotype prior. Its three values are normalised,

$$P_{ij}(g) = \frac{\Pr(a_{ij} \mid d_{ij}, g)}{\sum_{g'=0}^{2} \Pr(a_{ij} \mid d_{ij}, g')},$$

and an unobserved cell is exactly neutral, with $P_{ij}(g) = 1/3$. The reads of every sample, including libraries flagged by the check below, enter calibration and the genotype likelihoods.

2.2.2 Site and library quality control

The site filter calibrates itself on each contig's own depth and allele-count distributions, without sample labels, pedigree, benchmark or simulated truth. It is on by default for VCF input and applies to contigs with at least 1,000 admitted SNPs; on smaller contigs every site is kept.

The rules use a per-site genotype classification: an expectation–maximisation fit over three free genotype frequencies f_{j0}, f_{j1}, f_{j2} (no Hardy–Weinberg constraint) under a fixed binomial classification model ($e = 0.01$, heterozygote ALT fraction 1/2). It yields expected wrong-allele reads W_j and depth D_j^{hom} in homozygotes, and expected ALT reads A_j and depth H_j in heterozygotes. They also use a "clean core": the mode of a smoothed histogram, with separate Gaussian-equivalent left and right widths from its half-maximum crossings. Its window is the region around the mode where the core's local posterior probability is at least 0.5, an empirical-null construction (Bradley Efron 2004).

Five rules can remove a site:

- **r1** removes a site whose total depth $D_j = \sum_i d_{ij}$ lies strictly outside the clean-core window of total depth, in either tail. The window is calibrated on the contig, not a fixed global depth threshold.
- **r2** removes sites above the clean-core window of $\log X_j$, where the depth-heterogeneity index is $X_j = (N-1)^{-1} \sum_i (d_{ij} - E_{ij})^2 / E_{ij}$. The expected depth is $E_{ij} = D_j \bar{d}_i / \sum_{i'} \bar{d}_{i'}$, with $\bar{d}_i$ sample i's mean depth over sites passing r1.
- **r3** fits a two-class mixture to W_j. Clean sites follow $\mathrm{Poisson}(\lambda_j + \gamma_j \varepsilon_0)$. Artefact sites follow a negative binomial with mean $\lambda_j + \gamma_j \varepsilon_1$ and shape ϕ. The term $\lambda_j + \gamma_j e$ is the first-order expectation of W_j at a clean site with error rate e, including reads from heterozygotes and from misclassified homozygotes. The rule is applied only when $\varepsilon_1 \geq 3\varepsilon_0$, and it removes sites whose posterior probability of being clean is below 0.5.
- **r4** fits a two-component beta-binomial mixture to (A_j, H_j), labels the majority component clean, and removes sites whose clean posterior is below 0.5. It does so only when the minority component separates in latent spread or in mean, and abstains when the labelling is ambiguous (the majority mean lies farther from 1/2 than the minority mean) or no start of the fit reaches its stationarity certificate.
- **r5** (informativeness) removes sites with $N_j^{\mathrm{obs}}(1 - \max_g f_{jg}) < 2$: fewer than two expected carriers of a non-major genotype among the N_j^{obs} samples observed at the site.

A per-library check then tests for a foreign minor-read component, using cells at which only that library carries the cohort-minor allele. On each contig, a likelihood ratio compares a null mixture of the library's own error rates, its private heterozygotes and region-specific local artefacts with an alternative that adds a binomial foreign component of read fraction α. A library whose statistic reaches a working, Bonferroni-type threshold is flagged on that contig. A flag records a pooled likelihood preference, not a

contamination diagnosis. A flagged library is treated as missing for founder construction on that contig and is left out of the recomputed r5 carrier count; every other stage still uses its reads.

Founder discovery and local selection fit alleles only at discovery sites, where the total depth across all samples is at least five reads. This is a floor on read support, not a required fraction of observed samples. The retained site axis is cut into consecutive blocks of 200 sites, regardless of physical spacing.

2.2.3 Genetic coordinates

A genetic map gives cumulative centimorgans at knots with strictly increasing positions and nondecreasing map values; it is never sorted or repaired. The map is linearly interpolated between knots and extended continuously beyond them, or replaced on chromosomes without a map, at a fallback rate (5 cM/Mb by default). Model distances include this extension. Measured coordinates count only length the supplied map supports, unless the fallback rate is declared to be a measured genome-wide average. Where a stage needs a recombination fraction, a distance of δ Morgans is converted with the Haldane function $r_H(\delta)$ defined above (Haldane 1919), always in the same numerically stable form.

2.2.4 Relatedness

Relatedness infers full-sib families and their parents from genotype likelihoods alone; no painting, pedigree, phase or cohort label enters. It reads the evidence of every configured contig, for all samples on the filtered sites, so it starts only once all evidence exists. Founder reconstruction does not depend on it; only pedigree inference consumes it.

Sites and frequencies. At each site an ALT frequency p_j is estimated under Hardy–Weinberg proportions by six expectation–maximisation steps from $1/2$, clipped to $[10^{-4}, 1-10^{-4}]$. Unobserved samples contribute their prior expectation. Family sites are those with $\min(p_j, 1-p_j) \geq 0.05$ that are observed in at least half of the samples, thinned to every 20th such site. The M family sites of all contigs are concatenated in configuration order. We write $\psi_j(g)$ for the Hardy–Weinberg genotype probabilities at p_j, and $T(g \mid g_u, g_v)$ for the Mendelian probability of offspring genotype g given parental genotypes g_u and g_v.

Composite family likelihood. For each sample, a child term is defined over the nine parental genotype pairs:

$$C_{ij}(g_u, g_v) = \log \sum_g \tilde{P}_{ij}(g)\, T(g \mid g_u, g_v), \qquad \tilde{P}_{ij}(g) = \frac{P_{ij}(g)}{\max_{g'} P_{ij}(g')},$$

with $C_{ij} \equiv 0$ for unobserved cells. A family F consists of members and two parent slots, u and v, which carry no sex. A slot is either latent, with law $\Pi_j = \psi_j$, or filled by a sequenced sample i'. In that case its law is i''s genotype posterior from its own reads, $\Pi_j(g) \propto \tilde{P}_{i'j}(g)\,\psi_j(g)$. The family log-likelihood is

$$\ell_F = \sum_{j=1}^{M} \log \sum_{g_u, g_v} \Pi_j^u(g_u)\, \Pi_j^v(g_v) \exp\Big(\sum_{i \in F} C_{ij}(g_u, g_v) \Big),$$

and the objective is $\mathcal{L} = \sum_F \ell_F$ over a partition of all samples into families. A sequenced parent's reads are thus counted once, in its own family; in the families it parents, its law is the genotype posterior that conditions on those reads. Families are treated as independent given these laws, so $\mathcal{L}$ is a composite likelihood (Varin, Reid, and Firth 2011), not a joint pedigree likelihood. Two families F and G with latent parents may instead share one latent parent, with half-sib link likelihood

$$\ell^{\mathrm{hs}}_{FG} = \sum_j \log \sum_{g_u} \psi_j(g_u) \Big[\sum_{g_v} \psi_j(g_v) e^{S_{Fj}(g_u, g_v)} \Big] \Big[\sum_{g_w} \psi_j(g_w) e^{S_{Gj}(g_u, g_w)} \Big],$$

where $S_{Fj} = \sum_{i \in F} C_{ij}$.

Search. The search starts from one-member families with latent parents. Agglomeration then repeatedly merges the pair of families whose full-sib merge gain is the largest, provided that gain is positive and strictly greater than the gain from linking the same pair as half-sibs. Sweeps follow, each with three steps:

1. **Parent assignment.** A free parent slot is offered its family's best sequenced non-member and each of the family's members, a member first moving to its best other placement. An assignment must raise $\mathcal{L}$ by more than $\log N$ and keep the parent graph acyclic.
2. **Single-sample moves.** A sample may move to its best admissible family or to a new family of its own.
3. **One merge.** At most one merge is made per sweep, between families with compatible sequenced parents. Two latent families merge only when the full-sib gain beats their half-sib link.

Two searches run from the same agglomerated start: a best-improvement search, which applies the single largest move over all samples, and a first-improvement search, which visits samples in a fixed content order. Both are run because the objective has many local optima and neither search reaches the better one on every design. The content-order result is kept only when its objective exceeds the other's by more than 10^{-6} nats. Each search stops after a sweep that changes nothing, or after 30 sweeps. Gains within 10^{-6} nats are tied, and a tie goes to a content key: the lexicographic rank of each sample's complete child-term vector, never its name or index. Both searches are bounded heuristics, not an exhaustive search over pedigrees.

Support and links. Support is counted on units of measured map length, cut from each chromosome into equal pieces of at least 1 Morgan. Fallback distance is not treated as measured support, so a chromosome without measured length forms one unit. A member's support is the fraction of units on which its own family is strictly its best placement. A sequenced parent's support is the fraction of units on which it beats a latent parent in the same slot. After the fit, pairs of entirely latent families whose half-sib link gain, $\ell^{\mathrm{hs}}_{FG} - \ell_F - \ell_G$, exceeds $\log N$ are reported as a diagnostic.

Optional pairwise IBD. On request, each sample pair is run through a three-state (IBD0, IBD1, IBD2) hidden Markov chain (Lander and Green 1987; Rabiner 1989) on every 10th retained site, with genotype-likelihood emissions at frequencies p_j (Thompson 1975) and state persistence $\exp(-4\delta)$ over a model-map distance of δ Morgans. It reports forward–backward (Baum et al. 1970) posterior state probabilities averaged over jointly observed sites, and likelihood ratios against chains without IBD0 or IBD2. This quadratic-cost report is off by default and does not feed the family fit.

2.3 Local founder discovery and panel selection

2.3.1 Local founder discovery

Discovery proposes, independently in each block of a contig, a panel of binary founder haplotypes and its founder count K. Contigs are divided into consecutive 200-site blocks, irrespective of physical spacing. Within a block, discovery uses the M discovery sites (total depth of at least five reads over all samples) and a founder-construction view of the evidence in which samples flagged by contamination quality control are unobserved. We write $p_{ij}(g)$ for the genotype likelihood $P(r_{ij}, a_{ij} \mid g)$ normalised over g; a cell without reads has $p_{ij}(g) = 1/3$ and contributes nothing to any cost below. Samples with at least one observed discovery site are *active* (N_a of them); a block without any yields no panel. Samples are processed in canonical identifier order; identifiers affect nothing else. K is bounded only by the identifiability ceiling $\min(2N_a, 2^M)$.

Fixed-K model. A panel $H = (h_{kj})$ has K distinct binary rows. Each sample takes one of $(K+1)(K+2)/2$ diploid states z: an unordered row pair (k, k'), a row with an *unknown copy* (k, U), or two unknown copies (U, U). An unknown copy is a chromosome that no row

explains; at each observed site it takes the allele that fits best, at a penalty of $\lambda = 0.5$ nats. The cost of state z at an observed cell is

$$c_{ij}(z) = \begin{cases} -\log p_{ij}(h_{kj} + h_{k'j}), & z = (k, k'), \\ \lambda + \min\limits_{g \in \{h_{kj},\, h_{kj}+1\}} [-\log p_{ij}(g)], & z = (k, U), \\ 2\lambda + \min\limits_{g} [-\log p_{ij}(g)], & z = (U, U), \end{cases}$$

and zero at unobserved cells. Sites are grouped into consecutive bins of ten with costs $C_{ib}(z) = \sum_{j \in b} c_{ij}(z)$, and the panel's negative log-likelihood is a switching Viterbi cost over the B bins of the block,

$$\mathrm{NLL}(H) = \sum_{i} \min_{z_1, \ldots, z_B} \Big[\sum_{b=1}^{B} C_{ib}(z_b) + \tau \sum_{b=2}^{B} \mathbb{1}(z_b \neq z_{b-1}) \Big],$$

with $\tau = 10$ nats per change of state between adjacent bins. Discovery uses no genetic map.

Fitting. A start panel is fitted by alternating two hard steps. The A step gives each sample the state of least whole-block cost $\sum_b C_{ib}(z)$. States within a tie band of that minimum are tied; the tied category with fewer unknown copies is preferred, and a sample tied among several states of that category is summarised by the multiset intersection of their founder pairs (a differing copy becomes unknown), its evidence being shared equally among the tied states. The data cannot tell which tied founder supplies a copy, so no tied founder is privileged. The H step then updates rows one at a time in decreasing order of copy usage, setting each allele h_{kj} to minimise the summed cost of the samples carrying row k, given their partners' current alleles, with every cell's cost capped at its (U, U) cost; an allele changes only when the two allele costs differ by at least a tie quantum. When rows tie in usage, both consistent update orders are run and both endpoints kept. A trajectory ends at a fixed point, after 50 iterations, or on entering a cycle, when it returns the cycle's lowest-NLL state; a fit whose rows coincide is refitted at the smaller K.

Founder count. NLL ranks panels of equal K; different K are compared by a regularised leave-one-sample-out ("cavity") predictive score. Each active sample i is held out in turn. Its state is predicted from a Dirichlet–multinomial prior over states with concentration $\alpha = 0.5$, base masses π^0_z (uniform over ordered founder pairs, with mass $\omega = 0.02$ on states containing an unknown copy, a quarter of it on (U, U)) and the other samples' tie-shared state counts $n_z^{(-i)}$. Founder alleles at each site are predicted from a pairwise Markov random field built from the other samples' assigned states (unary terms from homozygous and row–unknown carriers, pairwise terms from heterozygous pairs, uniform prior), approximated by mean field (Jordan et al. 1999). The score is

$$\mathcal{L}^{\mathrm{cav}}(H) = \sum_{i} \log \sum_{z} \frac{n_z^{(-i)} + \alpha \pi^0_z}{N_a - 1 + \alpha} \prod_{j} \bar{p}_{ij}^{(-i)}(z) \; - \; \log [K(K+1)] \; - \; \log \binom{2^M}{K},$$

where $\bar{p}_{ij}^{(-i)}(z)$ is the expected genotype likelihood of state z under the mean-field allele marginals, averaged over the best converged mean-field optima; an unknown copy contributes both alleles with probability one half. The second term is the proper prior $P(K) = 1/[K(K+1)]$, and the third is the code length of an unordered set of K distinct binary rows of length M. The panel and assignments are fitted from all samples, including the held-out one, so $\mathcal{L}^{\mathrm{cav}}$ is an uncalibrated pseudo-score.

Search. Discovery explores complete panels by a bounded, reversible neighbourhood search. Starts at $K = 1$ are rows pooled from density-based clusters of samples (HDBSCAN* on expected genotype agreement (Campello, Moulavi, and Sander 2013)). Moves are: deleting a row; gauge rewires, which flip the alleles of both sides of a maximum cut of the graph linking rows co-assigned as heterozygous pairs, at sites where each side is internally constant and the sides differ, a flip that leaves the dosage of every sample whose copies straddle the cut unchanged; data births, which add a row seeded from the samples with the largest excess NLL over their per-site best genotype; soft-

residual births, which add consensus rows of clustered residual haplotypes left after subtracting each sample's well-supported founder copies; birth–delete replacements at fixed K; and, when residual births do not improve the selection, births from the best $(K+1)$-founder panels to $K+2$. Every proposal is refitted. A per-K archive retains leading NLL tie classes, and the leading class of each K is scored by $\mathcal{L}^{\mathrm{cav}}$. The selected set is the best tie class of $\mathcal{L}^{\mathrm{cav}}$ across K, then the smaller K. The search expands the next unexpanded NLL class of the selected K, ordinary moves before residual births, until no parent remains or a K-scaled budget is reached. The result is a local optimum of the generated neighbourhood: a block is flagged *certified* only when its winners' neighbourhoods were generated and scored completely without improvement, and *incomplete* when a budget omitted a whole tie class.

Release. The selected rows are latent: they, and any equally scored panels of the same K, start local selection. Discovery also releases alleles: the carriers' capped cost differences between alleles 0 and 1 of row k at site j are summed into log pseudo-odds, and an allele is called when its logistic pseudo-probability reaches 0.85 with at least two informative carriers; otherwise it stays unknown rather than adopting the latent value.

2.3.2 Local panel selection

Selection refits each block's panel under a haplotype-copying model and chooses it by a regularised objective, in an initial round and two rounds that use assembled founders as context.

Copying model. Each of a sample's two homologues copies, at each site, one of $S = K+1$ states: a founder row, or an unknown state whose alleles are independent Bernoulli(1/2) draws at each site (not a shared unknown founder). The state prior is $\pi = ((1-u)f_1, \ldots, (1-u)f_K, u)$ with founder frequencies f and fixed unknown mass $u = 0.01$. Between discovery sites $j-1$ and j, separated by x_j Morgans on the genetic map, each homologue independently redraws its state from π with probability $\epsilon_j = 1 - e^{-\gamma x_j}$ ($\gamma = 3$ per Morgan; a redraw may return the same state):

$$T_j(s,t) = (1-\epsilon_j)\,\mathbb{1}(s=t) + \epsilon_j \pi_t .$$

The first site draws from π. A state pair (s,t) emits $p_{ij}(h_{sj}+h_{tj})$ for two founders, $\frac{1}{2}[p_{ij}(h_{sj})+p_{ij}(h_{sj}+1)]$ for a founder and the unknown state, $\frac{1}{4}p_{ij}(0)+\frac{1}{2}p_{ij}(1)+\frac{1}{4}p_{ij}(2)$ for two unknown states, and 1 at unobserved cells. This is a diploid copying model (Li and Stephens 2003) with founders as templates; $L(H,f)$ and posterior expectations come from the forward–backward algorithm (Baum et al. 1970; Rabiner 1989) over ordered state pairs, at $O(S^2)$ cost per sample and site since homologues redraw independently.

Objective. Panels are compared with

$$J(H,f) = \log L(H,f) - D(H) - \log\left[K(K+1)\right] + \frac{1}{K}\sum_{k=1}^{K} \log\left(Kf_k\right),$$

whose last term shrinks frequencies towards uniform and is zero there. Without context the dictionary code is $D(H) = \log\binom{2^M}{K}$, as in discovery, so panels of different K are compared by the full objective, never by likelihood alone. In a context round, a row may instead be coded by reference to an assembled context founder ϕ plus its edits, at cost $\log V(n_\phi, d)$, where d counts the sites, among ϕ's n_ϕ called sites, at which the row differs, and $V(n,d) = \sum_{t \le d}\binom{n}{t}$. The panel's code is its shortest description,

$$D_{\mathscr{C}}(H) = \min_m \left[\Gamma_m(H) + \log\binom{2^M}{K-m}\right],$$

where Γ_m is the least total edit code over matchings of m context founders to rows, obtained for every m by successive shortest augmenting paths (Edmonds and Karp 1972). J is a regularised best-panel criterion, not a posterior over K, and its context code covers only called context sites.

Fixed-K fit. At fixed K, (H, f) is fitted by generalised EM (Dempster, Laird, and Rubin 1977). Each update takes a frequency step, $f_k \propto \bar{n}_k + 1/K$, where $\bar{n}_k$ is the expected number of initial and redraw-destination copies of founder k, the maximiser of the expected complete-data objective including the shrinkage term, kept unless J falls by more than 10^{-6}. It then flips, in the single row with the largest total, every site whose expected complete-data emission gain is positive. At a stationary point of these steps, the best exact one-bit flip is taken instead, with gain $\Delta_{kj} = \log L(H^{(kj)}, f) - \log L(H, f)$, where $H^{(kj)}$ is H with h_{kj} flipped. The fit stops when no flip improves, when the best moves tie between different panels (keeping the current panel), or when a flip would make two rows equal, since changing K belongs to the outer search. In context rounds the EM climbs the context-free J; the state along the trajectory that is best under the context-coded objective is returned if it beats the endpoint by more than the decision tolerance $\eta = 2^{-10}$ nats, and one-bit moves towards matched context founders are then taken while the context-coded objective rises.

Search. From a fitted start, each round proposes deleting any row, merging any two rows into an occupancy-weighted consensus row, and adding any row of a candidate bank. Deletions and additions are screened by their exact objective at the proposed frequencies, merges by an EM lower bound from the corresponding deletion's posterior, with up to $2K$ merges scored exactly. For each kind, whole tie classes of screened values are refitted until at least eight are taken. The best refit (best tie class, then fewer founders) replaces the current panel only if it improves the objective by more than η; otherwise the climb stops.

Rounds and context. The initial round starts from discovery's latent panel and its tied alternatives, whose rows form the bank. Before each of the two later rounds, the previous round's panels are assembled to the first, then the second, assembly level. Each component that exactly covers a set of blocks is refitted to their evidence by a block-level copying model whose weights for each block exclude that block's own emission, and a founder allele is released at site posterior 0.99 with support of at least two carriers. The refitted founders become the context and a proposal. The bank of a context round is built by refitting, with discovery's fixed-K model, the discovery panel and its tied alternatives, earlier rounds' released panels and the context proposal (missing alleles completed from the nearest latent rows). Each source's best panels under the cavity score become starts, and a neighbourhood search over bank rows, scored by $2\,\mathrm{NLL} + K\kappa(M/200)N_a$ with $\kappa = 0.5$ in the form of the Bayesian information criterion (Schwarz 1978), adds its selected rows. The distinct starts in the best class of fitted objective, with fewest founders, are each searched. The best outcome replaces the previous round's panel only if it improves the context-coded objective by more than η.

Release. An allele is called at the panel value when its one-bit conditional probability $1/(1 + e^{\Delta_{kj}})$ reaches 0.85 and favourable carriers support it. A favourable carrier is a sample whose own evidence favours the allele by more than 2^{-14} nats; support is their number with carrier probability above one half or their summed carrier probability, whichever is larger, and must reach two. A cell that passes the probability gate but lacks support is still released when the context founder matched to its row carries the same allele and the context refit released it. Distinct panels tied in the outcome are released conservatively: only rows present in all of them, called only where all agree. Each sample's largest number of unknown copies at any site is passed to assembly.

Both stages are bounded local searches with whole-tie-class budgets; neither establishes global optimality, and every budget omission is recorded.

2.4 Assembly of chromosome-scale founders

Local panel selection supplies, for every marker block b, a panel of R_b local founder rows whose alleles $y_{b\ell j}$ are either called (0 or 1) or explicitly unknown. A chromosome-scale founder k is a path $\pi_k = (\pi_{k1}, \ldots, \pi_{kB})$ that copies local row π_{kb} in block b, so its allele at a

site j of block b is $h_{kj} = y_{b\pi_{kb}j}$. A panel $\mathscr{P} = \{\pi_1, \ldots, \pi_K\}$ of paths spans a *component* of consecutive blocks. No sample label or metadata enters assembly.

2.4.1 Completion of unknown founder alleles

Before the hierarchy, a leave-bin-out ("cavity") posterior proposes fills for each block's unknown alleles. Sample diplotype weights are estimated from sites outside a 50-SNP focal bin at which every founder is called. At each focal site, the block's unknown alleles are enumerated jointly under Bernoulli(1/2) priors, exactly up to ten unknown founders and approximately above that. A fill becomes a candidate when its posterior reaches 0.99, at least two carriers support it, and its founder is distinguishable from every other founder outside the bin. An exchangeable founder therefore cannot acquire a fill from the bin alone.

An occupancy gate releases candidates only for founder rows that are demonstrably carried, judged from held-out carrier profiles of a two-fold cross-fitted variational EM over diplotypes and missing alleles (Dempster, Laird, and Rubin 1977; Jordan et al. 1999). A row qualifies through many confident carriers in its block, or through at least two samples that carry both the row and a uniquely matched row of an adjacent block. Folds are fixed by a deterministic hash of each sample's observation pattern, and each fold is profiled by a model fitted to the other, so no carrier profile uses the sample's own evidence. Released fills are published alleles only: every likelihood of assembly, refinement and painting, and every row distance of the count rule, reads the pre-fill calls, so a fill never feeds back into its own evidence.

2.4.2 Hierarchical joining

Assembly starts with one component per block and proceeds by levels. At each level, adjacent components are grouped into batches of at most ten, and each batch with two or more components is joined into one. A boundary is made a phase break, which is never joined, when fewer than four samples are informative on both sides. A sample is informative for a component if its genotype likelihoods are non-uniform at one of the component's linking sites. Batch sizes within a break-free segment form a palindrome, so the partition is orientation-symmetric. A batch is scored only on samples informative in all its components; elsewhere a sample's state would be arbitrary.

Levels stop when one component remains, when a level after the first fails to reduce the number of components, or when only phase breaks separate components after at least four levels. After every level, each component is refined (below). Founder additions in refinement are enabled only at the terminal level, so an earlier level's founder count is the working set of the next join.

2.4.3 Join objective

For sample i, site j and an unordered pair $\{k, l\}$ of panel paths, the emission is the robust, centred log predictive likelihood

$$\epsilon_{ij}(k,l) = \max\{\log(0.99 \sum_g \bar{p}_{ij}(g) \Pr(g \mid h_{kj}, h_{lj}) + \frac{0.01}{3}),\ -2\} - \log\frac{1}{3},$$

where $\bar{p}_{ij}(g) = P(r_{ij}, a_{ij} \mid g) / \sum_{g'} P(r_{ij}, a_{ij} \mid g')$. $\Pr(g \mid h_{kj}, h_{lj})$ is a point mass at $h_{kj} + h_{lj}$ when both alleles are called. An unknown allele is treated as Bernoulli(1/2), and two copies of one unknown local row share one latent allele. Contamination and floor bound any single genotype's influence. Emissions are summed over the usable sites (kept sites with a called founder) of bins $t = 1, \ldots, T$ that never cross a block, giving $E_{it}(k,l)$. Each input component gets about 20 bins of 10–100 sites, with at most 2,000 bins per batch. The per-sample score is a Viterbi maximum over sequences of unordered pairs (Rabiner 1989),

$$V_i(\mathscr{P}) = \max_{z_1, \ldots, z_T} [\sum_{t=1}^{T} E_{it}(z_t) - \lambda \#\{t > 1 : z_t \neq z_{t-1}\}],$$

with a free start and a uniform penalty λ per state change. λ is 20 nats up to 5,000 spanned sites and $20\sqrt{L_{\text{span}}/2000}$ beyond that, where L_{span} is the number of sites the batch spans. The join log likelihood $\mathrm{LL}(\mathscr{P}) = \sum_{i \in \mathcal{I}} V_i(\mathscr{P})$ sums over the $\tilde{N}$ informative samples $\mathcal{I}$ of the batch. LL compares equal-size panels; the count rule compares sizes.

2.4.4 Candidate paths and panel search

Candidate paths come from a linker. For every gap δ up to a horizon G, it fits haploid founder-transition matrices between components b and $b+\delta$ by EM (Baum et al. 1970; Dempster, Laird, and Rubin 1977). Homologues move independently between components; within a component, at most one homologue switches between adjacent sites, and a two-state error-tract process absorbs short discordant runs. G covers every gap at level one. At later levels, $G = \max\{1, 1 + \lfloor 0.5/(3\bar{D}) \rfloor\}$, where $\bar{D}$ is the mean component span in Morgans. An endpoint beam scores a path by summing, over components, the mean fitted log transition into the component from earlier components within the horizon. It keeps 16 paths per endpoint founder. The linker and beam only propose; the horizon never censors a term of LL.

The panel search starts from the best "anchor" panel, formed for one block from the first beam path through each of that block's rows. It alternates same-count moves with count steps. Tier 1 ranks bounded families of edits by fixed-painting gains: replacement by a pool path, local row changes, one-sided copies, suffix splices and cyclic exchanges. It scores the leading edits of each kind exactly and accepts the best improvement. At a tier-1 stall the count step runs; once it is quiescent, tier 2 bounds every move of an explicit neighbourhood and scores the survivors exactly, best bound first.

A same-count move must leave every changed path with a private cell (a block row no other path uses) and more than 1% mismatch to every other path. Otherwise the move would create a founder that the count rule deletes without having decided it. Finally, the beam is rerun with doubled endpoint quotas, up to 256 paths, or 1,024 if a move was accepted at 256. A panel is certified when no quota yields an improving move or an accepted birth. Certification is local to this neighbourhood and pool, not global.

2.4.5 Founder-count criterion

One rule decides every founder count in joins and refinement. Let $\mathscr{P}^+$ and $\mathscr{P}^-$ differ by one founder f, and let d_i be the per-sample gain of $\mathscr{P}^+$ under V_i (joins) or F_i (refinement, below). Each founder costs $c = \gamma \tilde{N} L_{\text{span}}/200$, with $\gamma = 0.5$ and $\tilde{N}$, L_{span} taken from the batch or component decided. A BIC-style rule (Schwarz 1978) would charge this against every sample, but a founder's evidence accrues only in the samples that carry it. The rule therefore charges $c/(2\tilde{N})$ per effective carrier, measured by the participation ratio of the positive gains (Kish 1965):

$$n_{\text{eff}}(d) = \frac{(\sum_i d_i^+)^2}{\sum_i (d_i^+)^2}, \qquad m(d) = \frac{\sum_i d_i}{c} - \frac{n_{\text{eff}}(d)}{2\tilde{N}},$$

where $d_i^+ = \max(d_i, 0)$. A gain vector passes when $m(d) > 0$ and $m'(d) > 0$, where m' omits the sample of largest gain ($\tilde{N}$ unchanged), so one large carrier cannot replace independent support. With $n_{\text{eff}} = \tilde{N}$, the margin is the BIC threshold $\sum_i d_i/c - \frac{1}{2}$.

The total gain alone does not suffice, because a recombinant of existing founders gains likelihood by saving switches without contributing new sequence. The rule therefore also tests the sequence part q_i: the score of $\mathscr{P}^+$ minus that of its shadow $\mathscr{P}^{+\sigma}$. The shadow replaces each block row of f that a reference panel lacks by the reference's nearest row there, using Hamming distance on co-called pre-fill sites. $\mathscr{P}^+$ is kept, or accepted, only if both d and q pass. A founder without a private cell is a pure mosaic: it has $q \equiv 0$ and is never kept. Equally near rows define a Cartesian family of shadows, evaluated completely up to 2,048 members. If the family's verdicts are opposed, the smaller count is chosen and the call is flagged ambiguous. A larger family is evaluated through its first member only, and the unchanged decision is flagged count-unverified. Each decision carries a

bootstrap band: individuals are resampled 2,000 times with a deterministic, content-keyed stream, and the 95% percentile interval of $\min\{m(d), m(q)\}$ is recorded (Bradley Efron and Tibshirani 1993). A band containing zero in its interior marks the call as ambiguous but never reverses it.

In a join, deaths precede births and the shadow reference is the smaller panel. Paths without novel sequence (no private cell, or within 1% mismatch of another path) are deleted unconditionally. Otherwise the death candidate with the largest $\mathrm{LL}(\mathscr{P} \setminus f)$ is tested. If that founder is kept, the novel pool path with the largest accepted gain is added; per-sample upper bounds first screen out paths that cannot pass. Pure-mosaic pool paths that would pass on total gain suggest that one panel path covers two founders' sequence; that path is copied, both copies are refitted, and the result is accepted if it survives its own death test and raises LL. Deleting a path born in the same search, or returning to an earlier count state, closes a count cycle: the deletion stands and births stop.

2.4.6 Founder refinement

Refinement re-optimises each component at fixed K, then revisits the count. Each homologue of sample i follows its own founder chain between consecutive informative sites, with transition $(1-\rho_{ij})\mathbf{I} + \rho_{ij}\mathbf{1}\mathbf{1}^{\top}/K$. The reset probability is $\rho_{ij} = 1 - \exp(-\Delta_{ij})$, where Δ_{ij} is the genetic distance in Morgans since the sample's previous informative site. Resets are thus Poisson along the map without interference, as in Haldane (1919), and a reset changes founder with probability $\rho_{ij}(1-1/K)$. Relative to staying, a change costs $\kappa_{ij} = \log\{1-\rho_{ij}(1-1/K)\} - \log(\rho_{ij}/K)$, and

$$F_i(\mathscr{P}) = v_i(K) + \max_{(k_j, l_j)_j} \sum_j [\epsilon_{ij}(k_j, l_j) - \kappa_{ij}(\mathbb{1}[k_j \neq k_{j-1}] + \mathbb{1}[l_j \neq l_{j-1}])]$$

is the maximum-path value of this copying model (Li and Stephens 2003), taken over every informative site without binning. The normaliser $v_i(K) = -2\log K + \sum_j 2\log\{1-\rho_{ij}(1-1/K)\}$ sums over the sample's informative sites after the first. It is panel-independent but depends on K, so it enters every comparison across sizes. Because both homologues share ρ_{ij}, the recursion reduces exactly to the $K(K+1)/2$ unordered pairs. $F = \sum_i F_i$ is a robust score, not a posterior probability.

At fixed K, each sweep has three steps:

1. Compute the panel's exact forward and backward messages.
2. Propose, for every founder and every window of eight blocks, the exact maximiser of a binned surrogate of F with the other founders fixed. From level two, windows of four first-level spans also propose whole first-level paths.
3. Walk the block junctions, scoring exactly every founder-pair suffix exchange and every proposed move, and accept those whose gain exceeds 2^{-9} nats.

Sweeps alternate direction until one accepts nothing, a local optimum of this move set; each fit runs from both directions and keeps the higher F. Moves spanning first-level spans are guarded. If such a move worsens the emission fit and raises F only through saved switches, it must still clear the threshold once the savings at crossovers shared between samples are removed. Descendants of one meiosis share its breakpoint, and without the guard a founder could absorb that crossover and become an artificial recombinant.

The count pass refits the incoming panel, then runs deletion rounds while $K > 2$. Each round refits the panel without each candidate founder (only those lacking a private cell, if any) and decides the best refit; the first rejection ends deletions. Addition rounds follow only at the terminal level. Each candidate is the panel plus a copy of one founder, refitted at $K+1$, and additions continue until one is rejected or K doubles. Refinement uses a symmetric null: the shadow reference is the larger panel minus each member of its own best unrefitted death class. The same two panels thus get the same decision whichever was the incumbent; disagreement among members selects the smaller K with ambiguity.

Joins treat paths within 1% mismatch as one sequence, whereas refinement, whose terminal decisions are final, scores the sequence part exactly. The decisions choose K, and within it the best converged fit of the pass is kept. The call is flagged ambiguous if that fit fails its own deletion test. A final polish (beam escapes, long repairs, exchange bridges) can raise F without changing K or revisiting the count.

2.4.7 Ties, alternatives, release and approximations

Continuous comparisons use tie classes whose width is a power-of-two quantum scaled to the score. Inside a class, joins consult linker evidence and then content order. A class that the data cannot separate is represented by the smaller founder count and then the fewest exact switches. Its other members are retained as families of alternatives. A later join evaluates every joint choice of its inputs' families, provided there are at most 16, on a common cohort and site set, and compares outcomes by LL within each founder count. At export, founder rows are cut wherever a retained family's panels disagree on continuity. Alleles are withheld at blocks where those panels disagree on content, and wherever no painted carrier has observed evidence. Count-unverified decisions and unperformed searches are recorded as metadata, never as biological alternatives.

With more than nine count candidates, only the nine best-ranked are refitted, so a successful candidate can be missed at large K. Shadow families and branches have explicit budgets, and a window search that exhausts its work account returns the component unchanged, recording the unperformed search. Unknown founder alleles enter every likelihood through a predictive approximation for each observation, not through joint integration across samples.

2.5 Painting, pedigree inference and founder refit

2.5.1 Painting samples as diploid founder mosaics

Each assembled component, a span of blocks sharing one founder panel, is painted for every sample as an independent hidden Markov model (HMM), so local ancestry restarts at every assembly break.

States and emissions. Painting uses the component's selected sites, those where at least one founder row is called. Rows whose calls (unknown calls included) agree at every selected site are indistinguishable by any sample and are pooled into one trajectory class. With $K' \le K$ classes, each homologue ("track") is in one class or in a BACKGROUND state that stands for a haplotype absent from the panel, giving $S = K' + 1 \le K + 1$ states. A class is anchored if it is called at some selected site. The ALT probability q_{uj} of state u at site j is the called allele h_{kj}, or, for an uncalled class and for BACKGROUND, $f_j = (1 + n_j^{\text{alt}})/(2 + n_j^{\text{call}})$, the Beta(1,1) posterior mean of the class calls at the site. Selected sites are split into $B = \max(1, \lfloor L_s/100 \rfloor)$ near-equal consecutive bins. With normalised genotype likelihoods $\tilde{\ell}_{ij}(g) = P(r_{ij}, a_{ij} \mid g) / \sum_{g'} P(r_{ij}, a_{ij} \mid g')$, the tempered bin emission of sample i in state pair (u,v) is

$$e_{ib}(u,v) = \frac{1}{D_{ib}} \sum_{j \in b,\ \text{observed}} \left[\max \left\{ \log\left((1-\epsilon)\sum_{g} \tilde{\ell}_{ij}(g) P_j(g \mid u,v) + \tfrac{\epsilon}{3}\right),\ \ell_{\min} \right\} - \log \tfrac{1}{3} \right], \qquad D_{ib} = 1 + (n_{ib} - 1)_+ \, r.$$

Here $P_j(g \mid u,v)$ is the genotype law of two independent alleles with ALT probabilities q_{uj} and q_{vj}, except that two tracks of one named class share a latent allele and so cannot be heterozygous. The genotype-error mixture is $\epsilon = 0.01$, the floor is $\ell_{\min} = -50$ nats, and n_{ib} counts the sample's informative sites in the bin (observed sites with non-flat likelihoods). The divisor D_{ib} is a design effect (Kish 1965). A product over sites treats them as independent given the copying pair, which overstates a bin's evidence, because panel errors run along rows and read errors are correlated between nearby sites. The correlation $r = 0.1$ is a constant tuned on simulations and sparse-read controls, not an estimate.

Copying chain. In the manner of the copying model of Li and Stephens (2003), each track's label jumps at rate ρ per Morgan, uniformly to one of the other $S-1$ labels. Between bin centres d_b Morgans apart, a track therefore changes label with probability

$$\theta_b = \frac{S-1}{S}[1 - \exp(-\frac{\rho d_b S}{S-1})],$$

floored at 10^{-15}, and the two tracks change independently. Each anchored class also has a panel-error state, which keeps the founder label but emits as BACKGROUND, representing a locally wrong row or a haplotype locally missing from the panel. While a label persists, its fidelity follows a two-state chain that enters error with probability α and leaves it with probability β. After a label change, fidelity is drawn from the chain's stationary law. The error state lets a track keep its founder label through a locally wrong row, paying an entry and an exit cost rather than two ancestry changes; painted switches count label changes only. For each component, (ρ, α, β) are estimated from all samples with an informative site by eight expectation–maximisation updates (Baum et al. 1970; Dempster, Laird, and Rubin 1977) from fixed starting values $(2, 10^{-3}, 0.25)$, each an exact scaled forward–backward pass (Rabiner 1989) followed by bounded maximisation. This is a deterministic, truncated estimator, not a test of convergence.

Decoding and release. Each sample is decoded by exact Viterbi recursions under two traceback conventions (the chain and its time reversal), which maximise the same score but break exact ties from opposite ends. Exchanging the two tracks is a gauge choice, not a switch. Where classes a and b have identical genotype laws at every site a sample observes over a run of bins, the data cannot place a switch from a to b within the run, so its bins carry the candidate set $\{a, b\}$ (a switch interval). Release is gated by two posterior masses with both tracks copying faithfully: q^F_{ib}, of the Viterbi unordered class pair, and q^{AF}_{ib}, of the corresponding pair of local allele-equivalence classes (classes with identical calls in the bin). Each mass is the smaller of its values for the two conventions' Viterbi pairs. With $\tau = 0.9$, a bin with $q^F \geq \tau$ releases both tracks as their classes. Otherwise, a bin qualifies when $q^{AF} \geq \tau$ and every track whose local class pools several classes is directly supported by an informative site where its class is called. Such tracks are then released as their local class, and the other track as its class. Otherwise both tracks abstain.

2.5.2 Label continuity

Where rows are locally interchangeable, assembly can let two founder rows exchange identity at a link point between blocks while the allele set at every site stays correct. Painting, however, reads label continuity as haplotype continuity. Every carrier of either haplotype then either switches at that point or exchanges its tracks, which flips a parent's phase frame and puts a false crossover into each child. Before the final decode, each component is therefore offered suffix transpositions, which exchange the labels of two singleton anchored classes from a link point onwards and so change emissions only in the bin straddling the link. Max-product messages of the label-level chain (without error states) give each sample's exact best-path gain for every transposition. The gains are summed over samples in units of one track change, and candidates gaining at least two units are ranked. The screen only ranks: each candidate is decided by re-decoding every sample under the full model. It is accepted when the total painted switch count falls by at least two and the samples whose count rises number at most 0.1 times those whose count falls. Under a genuine label exchange, the only losers are carriers with a real haplotype change inside the link interval. Under a real crossover inherited by one lineage, every other carrier loses a switch, so the ratio rule keeps such crossovers in the rows, even when the recombinant lineage is the majority. The search repeats until no candidate is acceptable, and the relabelled panel is kept only if repainting it strictly lowers the switch total. No pedigree, label or truth enters.

2.5.3 Pedigree inference

Candidates. Every sequenced sample is a child target, and every other sample is a candidate parent. Sample names are identifiers; no generation, cohort, sex or breeding metadata enters. A child's configuration is M0 (no observed parent), M1 (one candidate p) or M2 (an unordered pair $\{p, p'\}$); parent slots carry no sex. A slot without an observed parent follows a null source representing an unsequenced or unidentified parent, not a named individual.

Transmission model. A sample's source chain is an HMM over its ordered diplotypes, S^2 states, with painting's tempered emissions and label-level transitions. Its exact posterior marginals are symmetrised over homologue order. A parent transmits one homologue per bin, through a selector that switches between bin centres with the Haldane recombination fraction $r_b = \frac{1}{2}(1 - e^{-2d_b})$ (Haldane 1919), as in the inheritance vectors of Lander and Green (1987). Because the exact transmitted-state law is an S^4 object for two parents, we project it onto a first-order chain that keeps the exact marginal $\pi_{pb}(s)$ of the founder state transmitted by candidate p in bin b. Between bins, the chain uses a maximum-entropy transition matrix matching the exact adjacent marginals and the exact probability that the transmitted state persists (Csiszár 1975). Next to a parent bin with no informative markers, where the painted path is only interpolated, each bridge T is replaced by $(1 - r_b)T + r_b U$, with U uniform. Named classes transmit their panel allele (or f_j where uncalled); only BACKGROUND is conditioned on the candidate's reads, because the panel leaves alleles uncalled where data are noisy and conditioning on equally noisy reads would penalise true parents. The null source of child c starts from, and redraws from, the frequency of transmitted founder states among the other available samples, with the child left out:

$$\phi_{cb}(s) = \frac{\sum_{i \neq c} A_i \, \pi_{ib}(s) + w_0 / S}{\sum_{i \neq c} A_i + w_0},$$

where $A_i = 1$ if sample i has evidence in the component and $w_0 = 1$. The null slot keeps its state across an edge with weight $\kappa_b = (1 - r_b)(1 - \frac{S}{S-1}\theta_b)$ and otherwise redraws from $\phi_{c,b+1}$.

Composite score. The child's state in a bin is the ordered pair of founder states transmitted into its two slots, each following a parent's projected chain or, when unobserved, the null source. Up to ten evenly spaced markers per painting bin are scored. With transmitted ALT probabilities x and y, a marker contributes

$$P_{cj}(x, y) = 3 \sum_{g} \tilde{\ell}_{cj}(g) P(g \mid x', y'), \qquad x' = \varepsilon_{\mathrm{m}} + (1 - 2\varepsilon_{\mathrm{m}})x,$$

with y' defined likewise and mismatch probability $\varepsilon_{\mathrm{m}} = 0.01$. Each configuration a is scored by two forward passes (Rabiner 1989). One uses full-strength emissions. The other is tempered, raising each bin's emission to $\eta_{cb} = \min(3, n)/n$, where n counts the child's markers with non-uniform likelihoods in the bin's information group (consecutive bins holding at least 100 markers). The chromosome score is the composite

$$\ell_{cz}(\mathsf{a}) = \tfrac{1}{2}[\log L_{cz}^{(\eta)}(\mathsf{a}) + \log L_{cz}^{(1)}(\mathsf{a})],$$

summed over components. It pools two views of correlated evidence and is not an independent-marker likelihood. Certified upper bounds decide where M1 is evaluated exactly. Candidates far below the child's best genome-wide M1 are not retained and share a floor profile. A pair of retained candidates enters the score-defined M2 panel when a valid upper bound on its M2 comes within 150 nats of the child's best known score, in the full fit or in any leave-one-chromosome-out (LOCO) fit. Pairs outside the panel contribute nothing, so M2 evidence is a lower bound.

Genome-wide decision. For chromosome multiplicities w_z, the genome evidence of a non-M0 configuration is mixed once with M0:

$$\mathcal{L}_c(\mathsf{a}) = \log\left[(1 - \gamma)\, e^{\lambda \sum_z w_z \, \ell_{cz}(\mathsf{a})} + \gamma \, e^{\lambda \sum_z w_z \, \ell_{cz}(\mathrm{M0})}\right], \qquad \gamma = 0.02,$$

so a configuration that fails on part of the genome cannot score more than $\log(1/\gamma)$ below M0. The scale $\lambda \le 1$ maximises the predictive probability of held-out chromosome folds and is applied only when an interior value below one gives positive gain; it is an evidence weight, not posterior calibration. A configuration is testable only when its edges (and an M2 pair jointly) are exposed, in posterior expectation, in at least one bin, in 10% of painted bins, and on chromosomes holding three map units (Morgans of measured map, at least one per chromosome). Otherwise it contributes M0 likelihood to its state and cannot be named.

Direction is inferred from the data: a parent's painted junctions are inherited along the homologue it transmits, so per chromosome we compute the probability that the parent's inherited, detected junctions do not exceed the child's count. The two directions' log ratio is mixed with a chromosome-level painting-failure probability of 0.05 and summed over chromosomes. Three composite family terms are then added to every configuration: a direction term, bounded by a pair-level failure probability of 0.02; a reciprocal factor that excludes p being c's parent while c is p's, solved by converged cavity messages (loopy belief propagation (Murphy, Weiss, and Jordan 1999)); and a factor against reverse two-edge ancestry paths, computed over each sample's 16 best configurations.

State evidence is the log mean likelihood over a state's configurations, using the full number of eligible configurations $(1, N-1, (N-1)(N-2)/2)$ plus a uniform state prior. The parent-count state is chosen first, then the identity within it. A winner within a numerical tie tolerance is unresolved. Local choices are made acyclic by a bounded greedy repair that only moves a child to an equal or lower parent count, so the graph never creates parent evidence. Support comes from refitting everything downstream of the chromosome scores (λ fixed) on 1,000 chromosome bootstrap resamples (B. Efron 1979) and on C LOCO fits.

Release. Tier B, the primary release, releases a parent count when its bootstrap support reaches 0.7 and its LOCO support 0.8, when the child's informative chromosomes (those on which its own evidence has a unique winner) hold three map units, and when an M0 call persists under alternative state priors $(0.5, 0.3, 0.2)$ and $(0.2, 0.3, 0.5)$. For M1 and M2, some local parent must also be stable. This withholds counts that persist while the winning candidate changes, as with relatives of an unsequenced parent. An exact configuration further requires the local and graph configurations to agree, configuration and parent support, and direction log odds of at least $\log 49$ on every edge. Tier A uses 0.95 throughout. Edges that are stable and directed on their own can be released as marginal edges. An unresolved identity is reported as a candidate set of the fewest configurations covering 95% of the within-state probability.

Sibling-as-parent guard and count release. Both steps use a full-marker Mendelian exclusion e-value (Shafer 2021; Vovk and Wang 2021), computed from genotype likelihoods at jointly observed sites and never tempered. A dyad is excluded when $\log E \ge \log(2N/\alpha)$, with $\alpha = 0.01$. A parent of c is a parent of every full sib of c, and so is Mendelian-compatible with each of them. A relative wrongly released as c's parent is expected to be excluded from most of c's other sibs. For each released Tier-B edge $p \to c$, the guard takes c's full-sib family from the relatedness product, without p, without members joined to the family by released edges, and, as a half-sib heuristic, without members with a different released parent. It withholds the edge when at least two members remain and a strict majority of them are excluded from p. Withdrawals iterate to a fixed point, are never reversed, and return the child to unresolved. An unsequenced shared co-parent can veto a true edge, so an absent edge is not evidence that the parent is absent. For children whose Tier-B count remains unresolved, a count of zero or one observed parent is released when, after removing candidates whose reverse direction is supported, at most one unexcluded candidate remains and it agrees with the supported and acyclic-view parents.

Each child's release is thus an exact configuration with zero, one or two observed parents; a stable count with a candidate set; marginal edges with count bounds; edges withheld for unresolved direction or by the guard; or an unresolved state. A complete acyclic best-guess view is reported but is not a release.

2.5.4 Founder allele refit

Refit reconsiders individual founder alleles with the released painting labels held fixed. At each site, samples whose two tracks are released as singleton rows $\{a, b\}$, and that are observed in the founder-construction view, are pooled into a table of untempered robust log-likelihoods, $\mathscr{T}_j(a,b,g) = \sum_i \log((1-\epsilon)\,\tilde{\ell}_{ij}(g) + \epsilon/3)$. The table gives exact gains, other rows held fixed, for three moves: a single-row flip; a swap of the different alleles of two co-painted rows, which corrects an allele carried by the wrong row; and a joint flip of their equal alleles. Each site runs three sweeps of coordinate ascent. A move is accepted only when it gains 5 nats for each allele it changes, a hysteresis that gives the assembled allele prior odds of about 150:1. Within a round, qualifying moves are judged against the round's starting state and applied in order of gain, provided they do not interact. Two interacting moves of one tie class, where applying one would stop the other from qualifying, are alternatives the data cannot separate. Their rows' alleles are released as unknown rather than decided by row order. After the sweeps, alleles the panel left unknown are filled when one allele gains 5 nats and at least two carriers' own reads favour it. Pedigree inference never reads the refit panel.

2.6 Family phase, recombination maps and released outputs

2.6.1 Family phase conditional on the inferred pedigree

Family phase re-estimates every sample's two founder-label paths jointly with its released relatives, and releases genotypes, haplotype phase, phase sets and one transmission track per parent-to-child edge (a *meiosis chain*). It conditions on frozen upstream products: the painting's bins, states, estimated label-switch probabilities and tempered emissions; the founder allele refit; the released (Tier B) parent edges; and the genetic map. It refits none of them, and neither family phase nor the recombination maps feed back into any earlier stage. Each assembled component is solved separately, so no phase is carried across a component boundary.

State space. Let $x_{i\sigma b} \in \{1, \ldots, S\}$ be the painting state ("label") of homologue $\sigma \in \{0, 1\}$ of sample i in painting bin $b = 1, \ldots, B$. The states are the painting's classes of founder rows that are identical over its selected sites, plus a BACKGROUND state ($S = K + 1$ when every row forms its own class). When the pedigree releases a parent p for slot σ of sample i, that homologue also carries a source $v_{i\sigma b} \in \{0, 1, \perp\}$, an inheritance indicator as in multipoint linkage analysis (Lander and Green 1987). For $v \in \{0, 1\}$ it copies parental homologue v, so $x_{i\sigma b} = x_{pvb}$; for $v = \perp$ ("not a copy") its label is free, absorbing painting or panel errors, conversion, mutation or a false edge. A homologue without a released parent is a free label path. Slots are conventions: slot 0 and slot 1 carry no sex or parental-role meaning, and no sample metadata enters the model.

Transitions. Between bins $b-1$ and b, whose centres (the mean positions of their sites) lie Δ_b bp and d_b Morgans apart (supplied map, or a fallback uniform rate), the model uses a label-switch probability θ_b^{ℓ}, taken from the painting's estimated copying process; a meiosis-switch probability given by the Haldane map function (Haldane 1919),

$$\theta_b^{m} = r_H(d_b), \qquad r_H(d) = \tfrac{1}{2}(1 - e^{-2d});$$

and not-a-copy entry and exit probabilities $\alpha_b = 1 - \exp(-\rho_\perp \Delta_b)$ and $\beta_b = 1 - \exp(-\Delta_b/\mu_\perp)$, where $\rho_\perp$ is an entry rate per bp and $\mu_\perp$ a mean run length in bp. A free label keeps its label with probability $1 - \theta_b^{\ell}$ and moves to each other label with probability $\theta_b^{\ell}/(S-1)$; we write this law $T_b^{\ell}(x \to x')$. A linked homologue moves on the augmented state $z \in \{0, 1\} \cup (\{\perp\} \times \{1, \ldots, S\})$ with

$$T_b^{\mathrm{link}}(z \to z') = \begin{cases} (1-\alpha_b)\,[(1-\theta_b^m)\,\mathbf{1}(z'=z) + \theta_b^m\,\mathbf{1}(z' \neq z)], & z, z' \in \{0,1\}, \\ \alpha_b/S, & z \in \{0,1\},\ z' = (\perp, x'), \\ (1-\beta_b)\,T_b^{\ell}(x \to x'), & z = (\perp, x),\ z' = (\perp, x'), \\ \beta_b/2, & z = (\perp, x),\ z' \in \{0,1\}, \end{cases}$$

and starts in either copy state with probability $(1-\omega)/2$, or in $(\perp, x)$ with probability ω/S, where $\omega = \rho_\perp \mu_\perp / (1 + \rho_\perp \mu_\perp)$ is the stationary not-a-copy occupancy. A homologue entering $\perp$ draws its label uniformly rather than from the copied label, so that changing a parent's labels never alters a descendant's transition terms; this keeps block updates exact. Across user-supplied protected intervals (reference-order breakpoints, and both ends of reversed segments) θ_b^m is set to 1/2, so transmission is unlinked there.

Objective. With $e_{ib}(x, y)$ the painting's binned log emission of the unordered label pair, the stage maximises

$$J = \sum_i \sum_b e_{ib}(x_{i0b}, x_{i1b}) + \sum_{(i,\sigma)\,\mathrm{free}} \sum_{b \geq 2} \log T_b^{\ell}(x_{i\sigma,b-1} \to x_{i\sigma b}) + \sum_{(i,\sigma)\,\mathrm{linked}} [\log v_0(z_{i\sigma 1}) + \sum_{b \geq 2} \log T_b^{\mathrm{link}}(z_{i\sigma,b-1} \to z_{i\sigma b})],$$

where v_0 is the initial law above. The emission is the painting's robust log-likelihood of the bin's reads, $P(r_{ij}, a_{ij} \mid g)$, under the genotype distribution implied by the two labels, divided by the painting's design-effect factor for correlated sites within a bin. J is therefore a tempered log joint posterior of all labels and sources, up to a constant, not an exact likelihood. The emissions use the painting's pre-refit allele model, so the allele refit does not change this ancestry objective.

Optimisation. J is maximised by exact block coordinate ascent, one block per sample. Sample i's block holds both slot paths and, when i has children, a binary gauge path ζ_{ib}; $\zeta_{ib} = 1$ reads every child's source for i flipped at bin b. One block thus contains every move that exchanges the names of i's homologues from some bin onwards without breaking its children's transmissions. With all other samples fixed, the block is a hidden Markov model with $n_0 n_1 G$ states ($n_\sigma = S$ for a free slot, $S + 2$ for a linked slot; $G = 2$ with children, otherwise 1). Its emissions add the emissions of every descendant whose labels copy i's homologues; its transitions add its children's chain transitions. The Viterbi recursion (Rabiner 1989) solves the block exactly.

The ascent starts from each sample's free-label Viterbi path on its own emissions; samples with a released parent are then re-solved ancestors first, so that parental origin orients them. Sweeps visit samples ancestors first and keep a block's new path only when it raises J by more than a small gain quantum, computed exactly from the terms of the samples it changes. The ascent is monotone and stops when a sweep changes nothing, or at a sweep cap, which is recorded as unsettled. A converged ascent need not reach the global optimum.

Block moves alone cannot fix one configuration. Where both homologues of a sample without a released parent change label at the same bin edge, its own unordered-pair emissions cannot tell which old label continues into which new one; only its children can. A child copying the arbitrary pairing absorbs it by switching its other homologue, which no single block can undo. A joint move therefore exchanges the parent's homologues from that edge onwards, propagates the copied labels and re-solves each child's block. It is kept only when J rises by more than $\log[\tau_s/(1-\tau_s)]$, the odds of the set-continuation threshold τ_s defined below; a pairing whose alternative lies within that margin is recorded as ambiguous and breaks phase sets. The margin is a working value against one optimised alternative, not a calibrated pairing posterior.

Arbitrary conventions (visit order within a pedigree depth, tie-scan direction, descendant order) can decide ties, so the optimisation is run twice, independently, under opposite conventions. Release requires the two solves to agree, so that released outputs do not depend on a convention.

Release of genotypes and phase. For each sample and solve, with all other samples fixed at that solve's optimum, the exact forward–backward recursion (Baum et al. 1970; Rabiner 1989) over the same block gives posterior probabilities $\pi_{ib}(x,y)$ of the ordered label pair. These posteriors are conditional on the rest of the pedigree's optimal configuration; they are not joint marginals. At a released site j in bin b (a refit panel site inside a painting bin), a named state's allele $\hat{h}_{xj}$ is known only when all rows of its class carry the same called refit allele; BACKGROUND has no allele. Posteriors of the ordered alleles (u_0, u_1) carried by homologues 0 and 1 are

$$P_{ij}(u_0,u_1) = \sum_{x,y:\ \hat{h}_{xj},\hat{h}_{yj}\ \text{known}} \pi_{ib}(x,y)\,\frac{w_{xyj}(u_0,u_1)\,P(r_{ij},a_{ij}\mid g=u_0+u_1)}{\sum_{u_0',u_1'} w_{xyj}(u_0',u_1')\,P(r_{ij},a_{ij}\mid g=u_0'+u_1')}, \qquad w_{xyj}(u_0,u_1) = (1-\delta) f_{xj}(u_0) f_{yj}(u_1)\,.$$

Here f_{xj} gives allele $\hat{h}_{xj}$ probability $1-d_{xj}$ and the other allele d_{xj}, where d_{xj} is an empirical-Bayes estimate of the panel allele's uncertainty from the other carriers' reads and a chromosome-wide prior, with the sample's own direct term removed. The mixture mass δ redraws the ordered pair from a uniform genotype ($U = 1/3, 1/6, 1/6, 1/3$ for $00, 01, 10, 11$), representing a label error shorter than a bin, conversion or mutation. Posterior mass on pairs with an unknown allele is dropped, not renormalised, so such pairs withhold calls. A genotype is called from the unanchored posterior when its probability reaches τ_g (default 0.9). A called heterozygote is phased when $\max(P_{ij}(0,1), P_{ij}(1,0)) \geq \tau_p$ (default 0.99), and is otherwise released unphased.

A sample with a released parent is oriented by parental origin: homologue σ is the one transmitted from its slot-σ parent. A sample without released parents but with children is oriented through its children's chains (the gauge). A sample with neither has no anchor beyond convention, so each partition (opened by a forced break or by the end of the previous set) is anchored at its first bin whose label pair is heterozygous with probability at least τ_p, by requiring homologue 0 to carry the smaller label index there. The posterior is recomputed under that constraint, and the set ends at the first later heterozygous bin whose orientation certainty falls below τ_s (default 0.999). Orientation certainty is the median, over the bin's sites called heterozygous with probability at least τ_g, of $\max(P_{01}, P_{10})/(P_{01}+P_{10})$, with $P_{u_0u_1} = P_{ij}(u_0,u_1)$. Uncertainty about which labels a sample carries does not end a set where the candidate labels carry the same alleles. The stricter $\tau_s > \tau_p$ reflects that a wrong continuation misphases the rest of a set, whereas a wrong site call misphases one site.

Phase sets also break where the data cannot fix relative phase. The first case is painting tie boundaries in samples without a released parent, where the painting's track order was a convention; a sample with children bridges such a break when its children-anchored orientation certainty reaches τ_s on both sides. The second is identity junctions. A founder row is a trajectory through assembled blocks, not a certified biological homologue, so two rows can exchange identity at a block junction. Carriers of one row then switch labels there, whereas carriers of both keep continuous labels whose content changes homologue, which their own emissions cannot reveal. Where the free or not-a-copy tracks of at least two samples switch between the same two labels, a sample carrying both labels on both sides has its phase join withheld, unless a linked homologue copies one parental homologue across the edge and the parent's own reads pin that homologue's content on both flanks. The third is ambiguous pairings, which break the parent's and its children's sets. Breaks from both solves are combined. A genotype is released where both solves call the same genotype, and a heterozygote is phased only where both phase it; a new set starts wherever the solves' relative orientation changes, and the default solve's allele order is released. A phase set asserts relative phase within itself only; the orientation of separate sets is arbitrary.

Meiosis chains. For each edge e from parent p to slot σ of child i, the block posterior gives, per bin, the probability $\kappa_{eb}(v)$ that the child's homologue copies parental homologue v. Chains reset at the component start, across protected intervals and at either individual's phase-set starts. Parental homologue order is slot order for a parent with a

released parent, and otherwise each solve's gauge, aligned between solves by the parent's labels within each reset segment. A source is released where both aligned solves give $\kappa_{eb}(v) \geq \tau_p$ for the same v, with support $\bar{\kappa}_{eb}$, the smaller of the two solves' maxima.

2.6.2 Conditional recombination maps

Recombination maps are estimated from the released meiosis chains, conditional on the fixed inferred pedigree, the final phase and the supplied genetic map. An optional table of reference-order breakpoints removes the sites of every painting bin overlapping a breakpoint interval from the map axis and splits the component there, because painting and phase are rigid within a bin; this prevents misassembly or inversions from appearing as crossover pile-ups.

For each edge, released chain bins are markers at bin centres c_m with sources s_m. Consecutive markers within the same reset-free stretch define an interval $(c_{m-1}, c_m]$ with parity $o_m = \mathbf{1}(s_m \neq s_{m-1})$. An odd interval is a called crossover, reported with the outer bounds of its two bins and a joint endpoint support $\max(0, \bar{\kappa}_{m-1} + \bar{\kappa}_m - 1)$. Unreleased bins between markers are bridged but unresolved, so an even number of crossovers between markers cannot be seen; a lone marker contributes nothing, and no interval crosses a reset. Under Haldane's no-interference model, the number ξ of crossovers in an interval of genetic length λ is Poisson with mean λ, and odd parity has probability $r_H(\lambda)$. Each interval therefore contributes the conditional expectation

$$\mathbb{E}[\xi \mid o = 0] = \lambda \tanh \lambda, \qquad \mathbb{E}[\xi \mid o = 1] = \lambda \coth \lambda \ (\to 1 \text{ as } \lambda \to 0),$$

with λ taken from the supplied map at the bin centres, or from a fallback uniform rate without a map. It is spread over the interval by genetic length (uniformly in bp without a map or genetic length), and the interval adds one meiosis of exposure per bp. Both expectations tend to λ as λ grows, and to the called parity as $\lambda \to 0$. The estimate thus follows the called transmissions where markers are dense and reverts to the supplied map across long uninformative intervals. Called parity enters as data; the estimate is not a posterior computed from the chain model.

Expected crossovers X_t and exposure W_t (meiosis-bp) are summed over edges in fixed-width map bins t, split at component and breakpoint boundaries. The rate is $10^8 X_t / W_t$ cM/Mb and is missing where $W_t = 0$: uninformative sequence is not evidence of absent recombination. Observed genetic length accumulates rate times covered bp over the union of informative intervals only, so it omits gaps. Called crossovers per unit of callable exposure are reported alongside. A report-only detector groups overlapping calls into loci and flags patterns of transmission change that ordinary recombination does not explain; it never masks or refits the map.

2.6.3 Released outputs

For each contig, export writes a founder VCF and a phased-sample VCF (Danecek et al. 2011) on the same rows: by default every input biallelic SNP that reached the site filter, removed sites listing their removing rules in FILTER. The founder file has one haploid column per founder row of each released assembly segment. Labels are local to a segment, so no column asserts identity across segments. Alleles that are unknown, ambiguous or without read support from a painted carrier, and positions outside the segment's span, are `.`; at sites removed only as uninformative, the strict read-count majority allele may fill unblocked rows. The sample file holds the family-phase calls as GT:PS in input sample order, without imputation. PS is the position of the phase set's first site; an unphased heterozygote is written `a/b` with missing PS without ending its set, and filtered sites are `./.`. Recombination tables report per-meiosis calls, exposure and map bins in the units above.

2.7 Simulation with recorded truth

Accuracy is measured only on simulated crosses, because the real cross has no individual-level trio truth. A simulation (`[input] kind = "simulation"`) generates allele depths from founder templates and a known pedigree; `haplotypes evaluate` compares the finished run with the recorded truth. The generating read model, crossover process and map are configured independently of the inference model and map. Inference receives only sample names and REF/ALT counts, as from a VCF; the truth, including cohort and pedigree descriptions, is read only by evaluation.

2.7.1 Founder templates

A contig's template gives marker positions and, for each of H founder haplotypes (H even and equal across contigs), a probability for each allele. The `haplotypes templates` command builds templates from an observed VCF TO FILL: state which VCF and contigs. It runs the pipeline's block discovery on fixed-error evidence (filters and read calibration disabled) in 100 kb windows at a 50 kb stride, treating uncalled alleles as neutral (0.5, 0.5). A simple linker chains windows into up to six contig-length paths by penalised dynamic programming over junction mismatch scores, merging overlaps by adding signed allele information. Templates thus have realistic marker density and haplotype structure but are approximations.

Founder alleles are the more probable template allele at each marker. Exact ties, including neutral cells, are resolved by a per-contig random stream, so no reference bias enters the truth. Consecutive haplotypes are paired into $H/2$ diploid founders.

2.7.2 Pedigrees and partial sequencing

The default design has successive non-overlapping cohorts (default sizes 20, 100 and 200): cohort 1 descends from the founders and each later cohort from the one before. Each child receives an ordered pair of distinct, sexless parents drawn uniformly from the previous generation, so sibships arise by chance. From a configured cohort onward (default 3), each child is, with probability `backcross_fraction` (default 0), instead a backcross of a uniformly chosen previous-cohort member with one of its own parents. All contigs share one pedigree. Alternatively, an explicit pedigree table gives each individual's two earlier parents and an observed flag; its founders take the template founders in order and may be sequenced, and its sex and generation columns are truth descriptions read by no inference step.

Partial sequencing follows pedigree construction: optionally only listed cohorts are eligible, and a uniformly random $\lfloor \theta n_e \rfloor$ of the n_e eligible individuals is kept in pedigree order (θ = `observed_fraction`, default 1; at least three must remain). In the cohort design founders are never sequenced, so cohort-1 individuals, like any child of an unsequenced parent, have missing parents.

2.7.3 Meiosis

Each gamete starts on a uniformly chosen parental homologue and switches at each crossover. Without a generating map, crossovers form a Poisson process along the physical coordinate at the fallback rate γ (default 5 cM/Mb, i.e. 5×10^{-8} Morgans per bp), drawn as exponential gaps rounded up to whole base pairs. With a generating map, the number of crossovers between the first and last marker is Poisson with mean equal to their map distance x in Morgans, and their positions are uniform in genetic coordinates and mapped back to base pairs; contigs absent from the map use γ. Both are no-interference processes, consistent with the Haldane map function (Haldane 1919). Mutations arise at $\mu = 10^{-10}$ per bp per meiosis and flip the allele of the next marker without changing ancestry.

2.7.4 Reads

Each sequenced sample i has depth factor $u_i = 1$ unless the between-sample coefficient of variation v is positive; then u_i is lognormal with mean one and coefficient of variation v, renormalised to average exactly one. Optional dropout zeroes one contiguous tract of $w = \lfloor \phi L \rfloor$ markers per sample at a uniform start (ϕ = `dropout_fraction`), with depth scaled by $1/(1 - w/L)$ to keep the mean depth λ (default 5). In the default independent-site model,

$$d_{ij} \sim \text{Poisson}(\frac{\lambda u_i}{1 - w/L}), \qquad a_{ij} \mid d_{ij} \sim \text{Binomial}(d_{ij}, p_{g_{ij}}),$$

where d_{ij} and a_{ij} are the total and alternate read counts of sample i at site j, g_{ij} the true dosage, $p_0 = \varepsilon$, $p_1 = b$ and $p_2 = 1 - \varepsilon$, with error rate ε (default 0.02) and heterozygote alternate-read probability b (default 0.5; other values simulate allelic imbalance). A positive read length ℓ selects a paired-end fragment model instead. Fragment starts form a per-sample Poisson process scaled by the expected bases covered per fragment, so each marker keeps the same Poisson depth marginal while nearby markers share reads. Insert lengths are normal (default mean 375 bp, sd 150 bp), rounded and clipped to [50 bp, mean + 6 sd]; each fragment copies one uniformly chosen homologue and is read over ℓ bp from each end, overlapping mates counted once. Allelic imbalance converts a read of the disfavoured allele at a heterozygous marker with probability $|2b - 1|$, and errors flip reads at homozygous markers with probability ε. In both models heterozygous reads carry no sequencing error and alternate-read probability b.

2.7.5 Recorded truth and random streams

Per contig, the simulator records the true founder haplotypes; the maternal and paternal alleles of every sequenced sample; ancestry chunks, maximal half-open intervals of constant maternal and paternal founder origin; and each sequenced child's gametes with parent, slot, cohort and crossover positions. Chunk bounds are stored as integers, truncating fractional crossover positions, so truth can disagree at a boundary marker. A manifest records the requested and realised seed, contig order, sequenced samples and the generated pedigree.

All randomness comes from named streams with fixed generator construction and draw order: per-contig founder streams; a master stream that seeds a Mersenne Twister structural generator (Matsumoto and Nishimura 1998) and, per cohort or pedigree depth and before any scheduling, draws one seed per contig and child; a separate observation stream; and one read seed per contig, drawn for every contig even on resume. NumPy streams use PCG64 (Harris et al. 2020; O'Neill 2014). A configuration and seed therefore give the same data at any worker count.

2.8 Evaluation against recorded truth

`haplotypes evaluate` scores contigs independently and pools counts and spans, recomputing ratios and N50 values from pooled totals; undefined ratios are null. It scores the founder product of any stage (discovery blocks, three local-selection rounds, assembly levels 1–4, refinement, painting, founder refit) on its original component and row axes, plus pedigree, family phase and recombination. Per-sample local-ancestry paintings and relatedness families are not scored against truth. The metrics are not interchangeable.

2.8.1 Founder panels

A founder-panel component c has K_c rows over L_c markers (calls 0, 1 or missing). For row k and true founder haplotype f, e_{kf} counts the called markers where they differ and m_k the row's missing calls. Nearest-founder errors, $\sum_k \min_f e_{kf}$, are reported per million called alleles. Duplicate rows can score well by this measure without recovering every founder, so a one-to-one assignment π minimising $\sum_k (e_{k\pi(k)} + m_k)$ is also solved by rectangular linear assignment (Virtanen et al. 2020), with rows in content order so that exact ties never depend on row labels. With $\mathcal{M}_c$ the matched rows, the truth-to-panel count

$$T_c = \sum_{k \in \mathcal{M}_c} (e_{k\pi(k)} + m_k) + (H - |\mathcal{M}_c|)L_c$$

charges every allele of an unmatched founder; markers outside every component add H missing alleles each. A segment-wise score labels each row by its minimum-cost path through the true founders, a mismatch costing 1 and a label switch $\kappa = 20$. Its mismatches and switches are reported, and nearest-founder errors minus segment mismatches give chimeric charges. A split-charged cost adds κ per row at each component start after the first, so cutting a component cannot lower it, unlike the component-wise counts.

2.8.2 Ancestry strata and identity continuity

From the ancestry chunks, c_{fj} counts the sequenced homologues carrying true founder haplotype f at site j. Matched alleles are reported by copy stratum, absent ($c_{fj} = 0$), single ($c_{fj} = 1$) or multiple ($c_{fj} \geq 2$); these strata describe representation in the sample, not identifiability. Identity continuity labels each row with a founder in fixed 100-marker truth windows where it clearly matches one (Supplementary Information, section S6). Consecutive labelled windows of a row assert continuity, correct when the labels agree, classified as adjacent or gap-spanning and by stratum. Coverage counts pairs of nearby windows where a founder is represented and distinguishable that a single row labels with that founder in both.

2.8.3 Released phase

Phase is scored per sample within released phase sets, maximal runs of the sample's sites inside one painting component not separated by its released phase breaks. Scored sites are correctly called true heterozygotes released phased. A scored site's orientation is flipped when its first released allele differs from the true maternal allele. Per set, site errors count the scored sites on the minority orientation; switch errors count orientation changes between consecutive scored sites; long switches count changes between consecutive orientation runs of at least 20 scored sites, ignoring shorter runs. Unscored sites do not split sets. Genotype errors (an incorrect allele sum) are counted separately, never as phase errors. The phased share is scored sites over correctly called true heterozygotes. N50 is taken over first-to-last scored-site spans; a truth-free count and N50 of released sets are reported alongside. Results are also grouped by each sample's number of released inferred parent edges (0, 1 or 2), not true parentage.

2.8.4 Pedigree

Released Tier-B calls are compared with the true parents restricted to sequenced samples. Edges are matched by sample name, ignoring slot order; an unsequenced true parent is not an observable edge. A configuration is exact when the called parent state (zero, one or two observed parents) and the parent set both equal the truth; an unresolved call is never exact. Correct, extra and missing parent–child edges are counted separately.

2.8.5 Recombination

Map calls are scored only on inferred parent–child edges that are true gametes, reporting true crossovers within the inferred informative spans, the stage's own expected crossover count and exposure (bp × meioses), the called intervals, and the fraction containing a true crossover under an open-left, closed-right rule, for called and for reported intervals, with median widths. Hit rates are read with widths, because wider intervals hit more often.

2.9 Implementation and computational design

2.9.1 Task-graph execution

The pipeline is a task graph; each task declares its key, records, importable action, resource model and optional contig affinity. Only model prerequisites are barriers: relatedness waits for every contig's evidence; a contig's reconstruction (discovery through painting and founder refit) needs only its own evidence and runs alongside

relatedness; pedigree needs relatedness and all paintings; family phase needs the pedigree, painting and refit; recombination follows phase. Nothing feeds back upstream. Results are placed by task key, not completion order.

One runtime owns processes, threads, memory and compilation: forkserver workers start after the numerical-library policy is fixed, one CPU budget bounds processes × threads, contig lanes lend idle cores only at task boundaries, and admission follows measured memory pressure. A worker failure stops the run, keeping completed publications. Compiled kernels use Numba (Lam, Pitrou, and Seibert 2015) on NumPy arrays (Harris et al. 2020), with fast-math disabled. Workflow steps (per-contig scatters and genome-wide gathers) are cuts of the same graph, usable from workflow managers such as Snakemake (Mölder et al. 2021).

2.9.2 Deterministic numerics

Scheduling may change when work runs, never what it computes: scientific functions receive explicit parameters, arrays, streams and storage capabilities, and read no environment variables, files, clocks or memory measurements. Sums whose order would follow worker counts, batch sizes or memory layout use a declared order, so results do not depend on thread count. Near-tie decisions sort values exactly and group them into tie classes by consecutive-gap single linkage with quantum

$$q = 2^{\lceil \log_2 \max(q_0,\ \alpha|s|)\rceil},$$

where s is the decision's scale and α and q_0 a relative factor and floor for that kind of quantity (for the shared kinds, q_0 is sixteen times its measured noise motion, rounded up to a power of two); some decisions use a fixed width. Within a class, order follows a secondary scientific key, then exact discrete content, never a floating value; classes wider than $4q$ keep value order. An improvement must exceed q, and threshold rules use the band $[t-q, t+q]$ around threshold t. An explicit reversal setting tests whether an arbitrary content convention moves a scientific claim. Where version 4.0 replaced several arithmetic forms, or an execution-dependent order, by one declared form, the change is one of 15 declared numerical normalisations (L1–L15, Supplementary Table S2); none changes a model, objective, threshold, candidate set, search order or budget.

2.9.3 Checkpoints and resume

Each scientific product is published atomically to an append-only container under a derivation identity binding consumed source content (not paths or timestamps), upstream identities, parameters, tie settings, code closure, codecs and numerical policy, but not resources or output paths. Resume reuses a product only under a matching identity and refuses a mismatch rather than overwriting it; completion markers never authorise reuse.

2.9.4 Complexity in the number of founders

The design goal is one pipeline usable up to $K = 30$ founder haplotypes (documented envelope: $N \le 2,500$ samples on bounded crops), with nothing exponential in K and no stage above $O(K^3)$. Exponential candidate families are capped by named budgets; every stage's work and storage are modelled algebraically at $K = 30$. Four decided exemptions remain: rare context-coded local-selection screening, $O(K^5)$ per round (0); fixed ranked-refit caps in assembly and refinement, which can omit candidates at larger K (0b); bounded discovery and selection searches, including discovery fitting $O(NLK^4)$ and the selection backstop $O(NLK^3(K+L))$ (0c); and optional pairwise IBD, $O(N^2L)$, off by default (0d). Four worst-case costs above cubic (assembly joins' structural screening, refinement window fixing, painting label-continuity checks and founder-refit ties) are carried exactly as documented exceptions (0e) and are scheduled for review.

2.9.5 Kernel scaling ladders

K-dependence was also measured with kernel ladders on direct synthetic inputs, single-threaded and timed after a discarded first call: K from 5 to 30 at $N = 200$, and N from 250 to 2,500 at $K = 15$. The fitted log–log exponents β_K and β_N are reported with standard

errors, and $\beta_K \leq 3.3$ is accepted as at most cubic. A slope covers only the probed kernel, not an adaptive search's worst case, which for exempt stages remains analytic. Infrastructure without an intrinsic K was probed on size proxies.

3 Results

3.1 Accuracy on simulated crosses with recorded truth

We measured accuracy on simulated crosses generated from founder templates with recorded truth (Methods), scoring each run with the evaluation described above. TO FILL: the simulation matrix: template source, contigs, founder haplotypes H, cohort sizes and designs (including backcrossing and explicit pedigrees), fraction of individuals sequenced, depth, depth variation, allelic imbalance, dropout, paired-end fragments, and seeds and replicates per cell.

Because every stage's founder product is scored on its own components, the accuracy of the founder haplotypes can be followed through the pipeline, from the local panels of discovery to the refitted chromosome-scale founders (Table 1). One-to-one errors, rather than nearest-founder errors, are the primary measure, because duplicated rows can match the truth closely without recovering every founder; the truth-to-panel count additionally charges founders that no row recovers.

Table 1. Accuracy of founder haplotypes through the pipeline on simulated crosses. TO FILL from `haplotypes evaluate` (founders.csv and founder_components.csv), pooled over contigs and replicates, per scenario.

Stage	One-to-one errors per 10^6 called alleles	Truth-to-panel errors and missing calls	Segment label switches	Wrong identity-continuity assertions
Discovery	·	·	·	·
Local selection (initial, L1, L2)	·	·	·	·
Assembly levels 1–4	·	·	·	·
Refinement	·	·	·	·
Painting	·	·	·	·
Founder refit	·	·	·	·

Phase was scored within released phase sets on correctly called true heterozygotes, separately for samples with zero, one and two released parent edges, because phase in samples without released parents rests on the children's chains or on convention (Table 2). Pedigree calls were compared with the true parents among the sequenced samples (Table 3); a parent that was not sequenced is not an observable edge, and an unresolved call is never counted as exact.

Table 2. Accuracy of released genotypes and phase on simulated crosses. TO FILL from the phase tables of `haplotypes evaluate`.

Samples (released parent edges)	Genotype errors per called genotype	Phased share of true heterozygotes	Switch errors per scored site	Long switches	Released phase-set N50
All	·	·	·	·	·
Two parents	·	·	·	·	·
One parent	·	·	·	·	·
No parent	·	·	·	·	·

Table 3. Accuracy of Tier-B pedigree calls on simulated crosses, against the true parents among the sequenced samples. TO FILL from the pedigree table of `haplotypes evaluate`.

Scenario	Exact parent configurations	Correct edges / true edges	Extra edges	Missing edges
TO FILL	·	·	·	·

TO FILL: recombination: the fraction of called and of reported crossover intervals that contain a true crossover, with their median widths, and true crossovers within the inferred informative exposure against the expected count, on inferred edges that are true gametes.

The evaluation has no measure of per-sample local-ancestry (painting) accuracy or of relatedness families against truth; painting is scored only through its relabelled founder panel, and families only through the pedigree they inform.

3.2 Analysis of the TroMau cichlid cross

The real dataset is a laboratory cross of Lake Malawi cichlids (Malinsky et al. 2018) between *Astatotilapia calliptera* and a *Tropheops* species TO FILL: species name. The input is a single multi-sample BCF of biallelic SNPs with per-sample allele depths, restricted to the 22 autosomes TO FILL: sample count, sequencing platform, mean and range of depth, number of SNPs per contig, and how the BCF was produced. The sequenced samples include four individuals of the founding generation of the cross, three *A. calliptera* females and one *Tropheops* male, together with offspring sampled as first and second laboratory generations. These labels describe how the animals were sampled; they were not used in the analysis, and no sequenced founding-generation individual is assumed to be a parent of any other sample. The genetic map was a uniform placeholder rate of 5 cM/Mb TO FILL: confirm, or name the map used, so recombination distances and map lengths below are in units of that placeholder unless stated otherwise. Because that rate is not declared to be a measured genome-wide average, it provides no measured map length: each chromosome counts as a single unit of map support in relatedness and in the pedigree's release rules.

Because the real cross has no individual-level truth for its pedigree, we report internal consistency, stability and explicit ambiguity rather than accuracy: candidate margins, informative-site counts, support across chromosomes, stability under resampling, and the numbers of missing-parent, ambiguous and unresolved outcomes.

> TO FILL from the 4.0 TroMau run of the validation matrix:
>
> - per contig: sites kept by the site filter, samples flagged by the contamination filter, number of blocks, and the released founder count *K*;
> - relatedness: number and sizes of full-sib families;
> - pedigree, per release tier (A, B, C): released parent edges; children with two, one and no released parents; ambiguous and unresolved parent states; distribution of resampling support and of the number of supporting chromosomes; candidate margins;
> - phase: phase-set lengths and the fraction of sites phased per sample;
> - recombination: crossovers per meiosis and map length per chromosome.

3.3 Scaling with the number of founders and samples

Every stage has a work and storage model, evaluated at K = 30, and every stage was timed on synthetic inputs over a ladder of founder counts (K from 5 to 30 at N = 200) and a ladder of sample counts (N from 250 to 2,500 at K = 15), on one thread, after a discarded first call. The fitted log–log slope of time against K or N, with its standard error, is shown for each of the 39 probed scientific kernels in Figure 2 (Supplementary Table S1 lists them with what each probe covers).

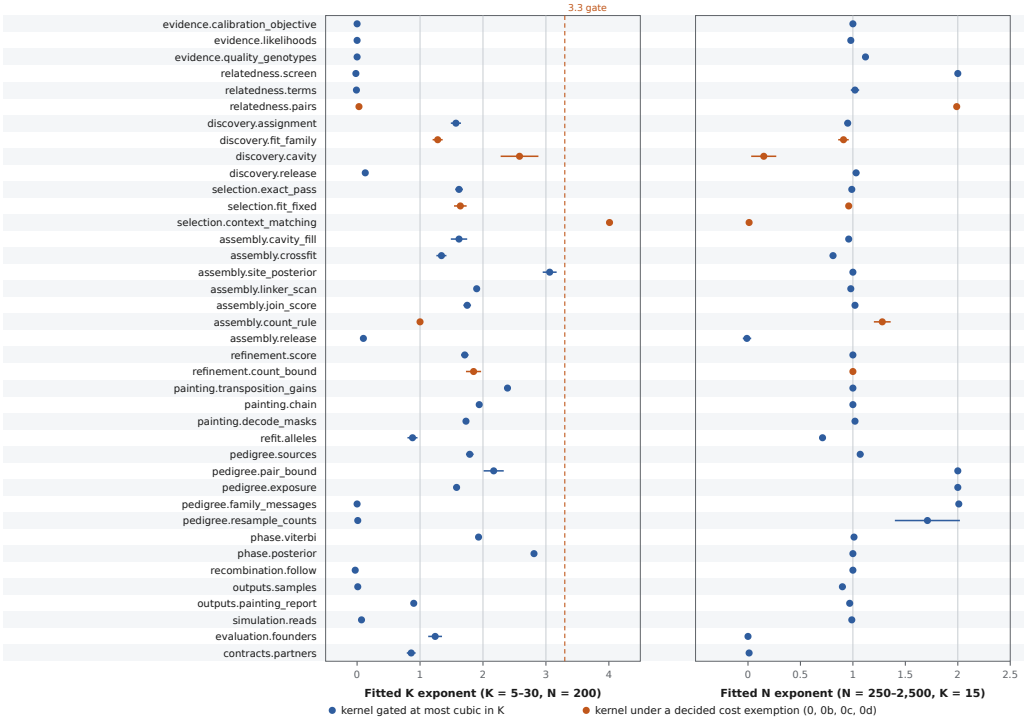


Figure 2. Measured scaling of the probed kernels of release 4.0. Each row is one kernel; points are fitted log–log slopes of run time against K (left; K = 5–30 at N = 200) and against N (right; N = 250–2,500 at K = 15), with bars of one standard error. A K slope at or below 3.3 (dashed) is accepted as at most cubic. Orange kernels belong to stages under a decided cost exemption, whose worst case is bounded analytically rather than by the probed slope. Five operational kernels whose size is only a proxy for K and N are omitted. Data: docs/developer/costs.md of the repository.

Every kernel gated as at most cubic in K had a measured K slope below the acceptance threshold of 3.3; the steepest were the posterior of the assembly site model (3.06 ± 0.11), the family-phase posterior (2.81 ± 0.02), the painting transposition gains (2.39 ± 0.05) and the pedigree pair bound (2.17 ± 0.16). The steepest kernel overall is the context matching of local selection (4.01 ± 0.05), which belongs to the rare context-coded screening covered by exemption 0, whose analytic cost is $O(K^5)$ per round. Of the 37 stage rows of the cost table, 21 are at most cubic in their probed kernels, 11 fall under a decided exemption or exception, and 5 have no intrinsic dependence on K.

Four mechanisms have worst-case costs above cubic in K that the probed kernels do not bound: structural screening in assembly joins, window fixing in refinement, label-continuity checks in painting, and tie resolution in founder refit. They are carried as documented exceptions; none of them binds on the validation inputs, and on those inputs founder refit used at most three rounds and at most two tie members.

Against the number of samples, most kernels scale linearly. Those with slopes near two are the stages that compare samples in pairs: the relatedness family screen (2.00 ± 0.00), the optional pairwise IBD output (1.99 ± 0.00; off by default) and three pedigree kernels (pair bound, exposure and family messages; 2.00–2.01). Quadratic cost in N is intrinsic to a model in which every sample is a candidate parent of every other.

TO FILL: wall time, CPU time and peak memory of the whole-genome TroMau run and of a simulated cross, on a stated node, from the 4.0 validation matrix.

3.4 Reproducibility and agreement with an independent implementation

The released outputs of the implementation do not depend on the number of cores: the TroMau cross gave identical outputs at 76 and at 56 cores. Release 4.0 is a reimplementation in which the models of the previous release (3.5) were carried over rule by rule into a new execution engine, and we used the earlier release as an independent

check. A matrix of 55 runs compared the two on simulated crosses with recorded truth, on the TroMau cross, on reference runs and on perturbed inputs. At the time of release, 30 runs had finished, all with exit status 0, and 29 had been scored TO FILL: final counts when the matrix completes. In every scored run the founders, paintings, founder refits, pedigree calls, phase and exported files matched the earlier release. The only differences were declared in advance as numerical normalisations (Supplementary Table S2): floating-point drift of at most 1.5×10^{-9} in pedigree diagnostics, canonical numbering of relatedness families, and diagnostic counts of rejected founder birth and death steps, which the new implementation no longer repeats (lower by 1 to 28 per run). None of these changes a released founder, painting, pedigree or phase call.

4 Discussion

We have described a method that reconstructs the founder haplotypes, pedigree, phase and recombination of an experimental cross from low-coverage sequencing, using only the allele depths of the sequenced samples and a genetic map. Its central choice is to let the data alone determine who is related to whom: no cohort, generation, sex or breeding record enters inference, every sequenced sample is a candidate parent of every other, and the direction of each relationship is inferred. The cost of that choice is that the method cannot be told which samples are founders or parents; the benefit is that errors in records cannot become apparent biological signal, and that the same pipeline applies to crosses whose records are incomplete or absent.

TO FILL: two or three paragraphs interpreting the simulation and TroMau results: where founder reconstruction, painting, pedigree and phase are accurate, how accuracy degrades with depth, missingness, backcrossing and unsequenced parents, and what the TroMau pedigree shows.

Several approximations should be kept in mind when interpreting the outputs. Discovery and assembly fit finite candidate panels with declared search orders and budgets, so a founder haplotype that no search reaches cannot be reported. Founder rows are inferred haplotypes within an assembled component; their labels are not chromosome-wide biological identities, and breaks, continuation ambiguity and identity flags qualify comparisons across components. The pedigree is chosen from a bounded candidate panel with composite evidence and chromosome-level calibration; its support fractions measure stability under resampling, not the enumeration of all joint pedigrees. Family phase releases modular calls conditional on the inferred pedigree rather than a single globally identified assignment of haplotypes. Recombination maps are conditional on the fixed inferred pedigree and the final phase, a supplied genetic map is a model input rather than observed truth, and an uninformative interval is not evidence for the absence of recombination. Because the method cannot exclude samples by label, individuals that do not belong to the cross must be removed from the input before analysis.

On the computational side, every stage is designed to be polynomial in the number of founder haplotypes, and the measured kernels of every stage gated as at most cubic in K stayed below that bound up to K = 30. Two kinds of cost remain outside that bound. Four decided exemptions keep bounded searches whose cost is documented and measured: rare context-coded screening in local selection, a fixed cap of nine ranked refits in assembly and refinement, bounded discovery and selection searches, and the optional pairwise IBD output. Four mechanisms have worst-case costs above cubic that are carried exactly as documented exceptions: structural screening in assembly joins, window fixing in refinement, label-continuity checks in painting and tie resolution in founder refit. None of them binds on the inputs analysed here, but they define where further algorithmic work is needed before crosses near K = 30 are analysed routinely; they are scheduled for review, as is the cap on ranked refits, which does not grow to enumerate every candidate at larger K. Independently of K, the pedigree is quadratic in the number of samples, because every sample is compared with every other as a candidate parent. The

documented scale envelope of N = 2,500 samples was established on bounded genome crops; it does not by itself establish that every whole-genome dataset will fit a given machine.

Finally, the implementation treats reproducibility as part of the method. Ties between numerically indistinguishable alternatives are resolved by declared content-based rules, random draws come from named streams with recorded seeds, sums are taken in declared orders, and released outputs do not depend on the number of cores. These properties made it possible to compare a complete reimplementation of the pipeline with its predecessor run by run, and to attribute every observed difference to a declared numerical normalisation.

5 Code and data availability

The software is available at https://github.com/HadiKhan-dev/fish_biallelic under the MIT licence. This manuscript describes release 4.0; release 4.0.1 adds the manuscript sources and leaves the scientific code of 4.0.0 unchanged. The repository contains the methods documentation of every stage (`docs/methods/`), the generated parameter and settings references, the cost and scaling tables summarised in Figure 2, the simulation and evaluation commands, and the sources and build script of this manuscript (`paper/`). The software runs on Linux x86_64 with Python 3.14 or later; exact tested dependency versions are pinned in `requirements-tested.txt`.

TO FILL: accession numbers and location of the TroMau sequencing data and BCF, of the simulation founder templates, and of the outputs of the validation matrix.

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